

The Long-Term Impact of Fertility on Cognitive Health: Son Preference, Missing Girls, and the One-Child Policy

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Abstract

Declining fertility and aging populations pose significant challenges for policymakers, yet the long-term effects of family size on parents' health and cognitive outcomes remain underexplored. Scientific evidence from epidemiologic literature highlights a u-shaped association between family size and cognitive health, but the causal impact remains unclear due to the endogenous nature of fertility choices and family formation. Using longitudinal household data from the China Health and Retirement Longitudinal Study (CHARLS), I estimate a causal effect of family size on parents' cognitive health at ages 45 and older by leveraging two sources of quasi-exogenous variation: the sex of the firstborn child and spatio-temporal variation in fines for unauthorized births during the One-Child Policy. Results indicate that the effect of higher fertility at the second margin on cognitive health is at best modest (-3.2% of a standard deviation). The coefficient is smaller than the OLS estimate (-5.7%), suggesting selection of cognitively endowed mothers into low fertility. Fixed-effects estimates suggest faster cognitive decline for mothers with two or more children with instrumentation amplifying, rather than attenuating, this finding. I additionally find that higher fertility substantially increases financial transfers received from children, suggesting that while a second child accelerates cognitive decline over time, compliers at this margin simultaneously benefit from increased material support in old age.

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1 Introduction

In recent decades, questions of how governments can increase fertility rates have gained prominence as policymakers confront aging populations and a declining workforce. Numerous countries have adopted pronatalist policies, ranging from cash transfers to subsidized childcare, to incentivize larger family sizes. Yet a critical barrier to designing such policies lies in understanding the long-term trade-offs faced by individuals: while having more children may result in more informal support in old age, it could also impose immediate financial and health burdens in younger years, which could have long-lasting health effects on parents of children. If elevated stress or reduced lifetime savings due to child-rearing costs negatively affect well-being in later life, policymakers may need to pair pronatalist measures with expanded public safety nets, such as pensions or healthcare subsidies, to achieve desired fertility outcomes without compromising elderly health.

This paper aims to establish a causal effect of fertility on cognitive health in old age by addressing the issue of endogenous fertility choices. Fertility decisions are inherently correlated with unobserved household and individual characteristics, such as preferences for health investments and preferences for children, which confound the estimated relationship between fertility and cognition. For instance, parents who invest more in their own health may also choose to have fewer children, creating a spurious correlation that can bias ordinary least squares coefficient estimates.

To isolate the effect of marginal fertility from selection effects and obtain a causal local average treatment effect (LATE), I leverage quasi-exogenous variation from two sources. First, I use the sex of the firstborn child as an instrument for family size, building on evidence that parents in certain cultural contexts including China are more likely to have additional children if their firstborn is female ([Ebenstein, 2010](#)). Second, I exploit geographic and temporal variation in fines imposed for exceeding fertility quotas under China’s One-Child Policy (OCP).

During the One-Child Policy, local governments implemented locally adjusted fine rates for unauthorized births after receiving approval from the Provincial Standing Committee of the People’s Congress. The fines could amount to 10–20% of both parents’ wages for a period of 3 to 14 years ([McElroy and Yang, 2000](#)). Enforcement varied between and within provinces, across years, ethnicity and local household registration (hukou). For example, the policy was stricter in urban areas, while mothers of daughters in certain rural areas were permitted to have a second, or even third child in some cases ([Li and Shi, 2025](#)). Ethnic minorities were officially exempt from the policy implementation of the OCP, but not unaffected as peer effects appear to have also shifted preferences toward a low-fertility equilibrium ([Xiao, 2024](#)).

To parametrically estimate the causal impact of family size on cognitive health, I use data from the China Health and Retirement Longitudinal Study (CHARLS), a nationally representative panel survey containing detailed information on demographic backgrounds, family composition, employment and mental health. The data contains a rich assessment of cognitive health among respondents. Both the main respondent and their partner are assessed on immediate and delayed word recall, the serial 7s subtraction test, and temporal orientation. These tests are well-established measures of assessing cognitive health and are thus widely considered to be a high-quality assessment of population-wide prevalence of cognitive health among elderly.¹

I merge this data to an external dataset on fines imposed for unauthorized births that was compiled by [Ebenstein \(2010\)](#), which contains provincial and temporal variation in penalties for unauthorized births under the One-Child Policy across provinces between 1979 and 2000. The level of these fines or their exemptions have been used extensively in recent years, mostly focusing on the widely debated quantity-quality tradeoff ([Guo et al., 2025](#); [Xiao, 2024](#)), the effect of the policies on fertility ([García, 2024](#)), or marriage outcomes in adulthood ([Huang et al., 2021, 2023](#)). The fines needed to be approved by the provincial government and were implemented by sub-provincial governments, and it is possible that these were set endogenously based on local fertility demand ([Li and Shi, 2025](#); [Zhang, 2017](#)). Because of this, I control for local demand for children through the inclusion of granular county-fixed effects (which is at the sub-prefectural level), which allows one to compare similar individuals exposed to the same fertility policies in the same regions during their childbearing years.

Family size can have direct implications on the cognition of elderly parents. For one, having more children may increase financial and time transfers from these children to elderly parents, which could help them remain cognitively intact. For another, having more children may also prevent them from attaining a high level of education because of resource constraints within the household during childhood, as noted in the discussion of the widely debated child quality-quantity tradeoff ([Becker and Lewis, 1973](#); [Black et al., 2005](#); [Guo et al., 2025](#); [Rosenzweig and Zhang, 2009](#)). If better-educated children invest more in elderly parents, fewer but better-educated children could improve parental cognition ([Ma, 2019](#)). The overall effect of family size on elderly cognition remains thus theoretically ambiguous and highly context-dependent. Recent causal approaches using the sex composition of the first two children as an instrument suggest that more children at the third parity causally worsen elderly cognition ([Bonsang and Skirbekk, 2022](#)), shorten lifespan of both

¹The sister studies Health and Retirement Study in the US (HRS), the English Longitudinal Study of Aging (ELSA), and the Survey of Health and Retirement in Europe (SHARE) contain similar and in some cases identical questions.

parents in the US (Fletcher and Noghanibehambari, 2025), while Fischer et al. (2025) do not find any such causal impact on dementia prevalence in Sweden using administrative data.

The question of whether having more children has a causal effect on cognitive health is especially relevant in the Chinese context. In rural areas across China, the elderly often continue working into old age or depend heavily on support from children due to a lack of a comprehensive pension system.² According to Rao and Zhang (2024), 41% of post-retirement individuals and 69% of elderly older than 70 rely on children for support in old-age. Children often provide not only financial assistance, but also emotional and practical care which is important for slowing cognitive aging. The Local Average Treatment effect of fertility on cognition may therefore differ in developing country settings.

My findings suggest that the local average treatment effect of having two or more children on the level of cognitive health is at best modest. The point estimate for the effect of fertility on standardized cognition indicates a reduction of 3.2 percent of a standard deviation in my preferred specification. Although the instrumental variable estimate is statistically insignificant, which might partly be due to a lack of statistical power, the magnitude suggests that if an effect exists, it is economically small. The causal estimate is even more modest than the uninstrumented OLS estimate of 5.7 percent, which is suggestive of selection into lower fertility among cognitively endowed mothers. Leveraging an individual fixed-effects specification to model cognitive trajectories, fixed-effects OLS estimates suggest that having two or more children accelerates the rate of cognitive decline. The fixed-effects IV estimates yield point estimates of a larger absolute magnitude. This amplification indicates that the uninstrumented FE-OLS estimates are biased toward zero due to attenuation from selection.

Regarding other outcomes that are related to elderly cognition, I find no causal impact of children on labor supply in old age, but higher fertility does causally increase the probability of receiving financial support from children as well as a higher monetary amount thereof. Overall, these findings suggest that while earlier studies linking higher fertility to worse baseline *levels* of cognition largely reflect positive selection into low fertility among cognitively endowed mothers, having a second child does causally accelerate the *trajectory* of cognitive decline over time, despite the increased material support in old age.

The remainder of this paper is organized as follows. Section 2 reviews relevant literature on the One-Child Policy and the relationship between fertility and parental outcomes. Section 3 introduces

²Some recent reforms have improved social security in rural areas of the country, such as the New Rural Pension Scheme introduced between 2009 and 2012, but payout amounts are relatively small compared to living costs in those areas.

the data and construction of variables. Section 4 outlines the conceptual framework and empirical model. Section 5 presents the estimation results. Finally, Section 6 concludes with a discussion of the findings and their implications.

2 Background and context

2.1 Review on effects of family size on cognition

Several studies document a relationship between fertility history and cognition, but it is unclear whether the relationship is causal ([Bonsang and Skirbekk, 2022](#)). Elderly who earlier in their life chose to have more children and thus a larger family size may be inherently different from those who chose to have fewer children. It is therefore difficult to separate selection effects from causal effects.

In a worldwide context, correlational evidence generally suggests a u-shaped relationship between fertility and late-life cognition, with worse cognitive health among individuals with no children and one child, and among individuals with more than three (or four) children, compared to those with two (or three) children. This relationship appears to be consistent across many different contexts and regions. For instance, [Read and Grundy \(2017\)](#) found in the English Longitudinal Study of Ageing that individuals with few (0/1) and many (3 or more) children have lower cognitive functioning compared to those having two children, but these associations are weakened after adjusting for socioeconomic factors. [Bordone and Weber \(2012\)](#) report that in the Survey of Health And Retirement in Europe (SHARE) childless older adults and those with large families had lower performance on fluency tests but not on recall-type tests. [Mekonnen et al. \(2025\)](#) also observed a u-shaped association using Norwegian registry data, with individuals having no children, one child, or four or more children showing higher dementia risk than those with two to three children. In Sweden, [Fischer et al. \(2025\)](#) identify a lower prevalence of dementia among both male and female parents of two (compared to less than or more than two) children (but they do not identify a causal relationship at the third parity).

To deal with selection, recent works aiming to study the impact of family size on outcomes later in life include various identification approaches. Following foundational work in causal inference exploiting child sex composition and twin births as instruments for family size ([Angrist and Evans, 1998](#); [Rosenzweig and Wolpin, 1980, 2000](#)), some research has adapted the methodology to relevant contexts. For instance, in the context of China, [Oliveira \(2016\)](#) exploits the randomness of twin

births as an IV for fertility in China to study the impact of children on financial transfers and coresidence. However, as noted by [Huang et al. \(2016\)](#), during the One-Child Policy many families tended to report children as twins to avoid fines associated with above-quota births, which would invalidate the exogeneity assumption of the twin-births instrument (census data may be especially at risk of providing unreliable information about a twin birth as individuals may purposely have registered different children as twins to avoid penalties). The exogeneity of dizygotic twin births has more generally been called into question as well, since the prevalence is correlated with maternal age ([Farbmacher et al., 2018](#)) and maternal health ([Bhalotra and Clarke, 2019](#)).

[Chen and Fang \(2021\)](#) exploits differential provincial exposure to the family-planning leading groups during the Long-Later-Fewer (LLF) campaign to find that fertility declines came with an increase in depression among elderly without compromising physical health. Research using the LLF as exogenous variation identifies effects from fertility declines at higher parities, since cohorts affected by the LLF completed their childbearing years before the introduction of the OCP. [Rao and Zhang \(2024\)](#) exploit the same variation along with firstborn gender to find that additional children reduce elderly labor supply at post-retirement ages, but do not identify any health effects. Related to this work, [Wang \(2025\)](#) also uses firstborn gender as instrument to conclude that having more children reduces labor supply among the elderly, but does not use policy exposure to the One-Child Policy. Reduced labor supply may be a mediating channel in the causal relationship between children and cognition ([Celidoni et al., 2017](#)).

Only a few causal studies examine the impact of fertility on cognition. [Bonsang and Skirbekk \(2022\)](#) use the sex composition of the first two children as instrument for having a third child in Europe using SHARE data from 2004 to 2016. This identification strategy is suitable in European contexts without fertility restrictions, because sex-selective abortion is relatively uncommon, whereas in China increased prevalence of ultrasound machines increased the incidence of sex-selective abortions during the 1980s. They find that at the third parity, higher fertility is related to worse late-life cognition for both men and women. The identification strategy in their study is equivalent to [Cáceres-Delpiano and Simonsen \(2012\)](#), who use the National Health Interview Survey 1982-2000 and find that a larger family size increases the likelihood of divorce, as well as the likelihood of high blood pressure and obesity.

[Islam and Smyth \(2015\)](#) and [Bai et al. \(2025\)](#) both use CHARLS data to investigate the impact of children on elderly health. [Islam and Smyth \(2015\)](#) instrument the number of children using a single interaction between firstborn gender, rural household registration and the policy rule. [Bai et al. \(2025\)](#) use an exposure-based identification approach. They exploit differential rollout timing

of LLF policy (which preceded the One-Child-Policy), and an OCP exposure measure based on policy types, but they do not use fines for unauthorized births. They find a negative relationship between both exposure measures and fertility, and a positive effect of fertility on cognition at low parities.

Naturally, the impact of fertility on cognition would depend on the selection effects that apply in the relevant contexts. Selection effects may also vary based on whether births were desired and planned, or whether instead the births were unwanted. The increased access of ultrasound in the 1980s in China may thus have lowered the share of unwanted births, as increased access to contraception has done in the US (Kearney and Levine, 2025). The Chinese context effectively offers a case where fertility was 'rationed' by policy, thus impacting individuals differently based on the policy environment they were exposed to at the time. A theoretical framework distinguishing wanted and unwanted births is formalized in Guo et al. (2025).

The contributions of the current paper are several. Most research linking fertility history to late-life cognition is of correlational nature (Gemmill and Weiss, 2022; Read and Grundy, 2017; Saenz et al., 2021). Causal evidence is limited, and the correct approach to study impacts of family size is, as I have discussed above, highly context-dependent. I establish plausibly causal estimates using my identification strategy because it does not rely on twin births. Using twin births as instrument may be inappropriate because the additional birth may have just been reported as such (Huang et al., 2016), and twinning may not be randomly assigned but related to unobserved characteristics of the parent (Bhalotra and Clarke, 2019).

Second, while extensively used in other research settings (García, 2024; Huang et al., 2021, 2023; Xiao, 2024), this paper is the first to use these fines for unauthorized births as plausibly exogenous variation of family size to study cognitive health at older ages. Solely using child gender or children sex composition may be insufficient to provide credible causal effects as marginal individuals may be difficult to identify (for instance, preferences for mixed-sex children may vary across different cohorts). This paper provides a more precise causal identification of the Local Average Treatment Effect precisely at the second-child margin. Rather than relying on broad cohort-level policy exposure, such as in Bai et al. (2025), I exploit individual-level quasi-exogenous variation using the sex of the firstborn child interacting with finely-measured temporal and provincial variation in financial penalties. By including granular regional fixed effects, I compare otherwise similar individuals within the same counties. Furthermore, I expand the focus by directly investigating the trajectory of cognitive decline over time and explicitly testing intergenerational transfers as a mediating mechanism. Among recent papers at the knowledge frontier of the conceptualization of

the exposome, i.e. the various environmental exposures over the life course that influence brain health and dementia outcomes, the current paper contributes to the strand of the literature that investigates how cognitive health develops in old-age as a function of endogenous fertility choices (Bai et al., 2025; Bonsang and Skirbekk, 2022; Fletcher and Noghanibehambari, 2025).

2.2 Evolution of the One-Child Policy

In the 1970s, China underwent a significant shift in its demographic policy. After two decades during which the government had been explicitly encouraging population growth, which resulted in doubling of the population from 540 million in 1949 in less than three decades, policymakers reversed course and enacted a series of measures designed to curb it. This new era of population control was formally defined in 1979 with the introduction of the One-Child Policy (OCP), which was the first time that family planning policy was formally encoded into Chinese law (Banister, 1987).

In principle, the OCP mandated that each couple could have only a single child (termed *dusheng zinu*). However, its practical implementation was far from uniform and varied significantly based on ethnic group and hukou type. The restrictions primarily focused on the Han people, the largest ethnic majority which constitutes approximately 92% of the population (Huang et al., 2021). Across most provinces, ethnic minority groups were either permitted to have a second or third child, or were exempt from restrictions altogether.

Urban and rural residents within the same provinces in general were also treated differently based on the hukou status of residents. In China, ‘hukou’ is a household registration system that essentially ties access of public services to the residency of parents. While the hukou system was introduced in 1958 during socialist reforms as a means to limit migration to cities, it continued to be implemented until recent years, albeit being partially liberalized after the economic reforms in the 1980s (Huang et al., 2024). The hukou system mandated that each person be registered in their place of origin.³ The OCP was most strictly enforced in cities, where urban hukou holders were almost universally limited to one child per couple. In rural areas, however, couples were typically allowed to have a second child if their firstborn was a girl—a common exception that led the policy to be known as a “one-and-a-half (or 1.5) children policy”.

Recognizing the vast and diverse demographic and socio-economic conditions across the country,

³While migrants were later permitted to relocate, access to public services (such as the education of children) in urban destinations remained contingent on local registration. This resulted in many migrant workers leaving children behind in the countryside to find higher wages in cities.

the central government began to decentralize policy authority already in the early years of the policy. In February 1982, it amended the constitution and issued “Document 11”, which empowered local governments to draft and implement specific regulations suited to local circumstances (Baochang et al., 2007; Huang et al., 2021). This move towards local flexibility was reinforced two years later with the issuance of Document 7, which stipulated that the responsibility of birth control should move from the central government to provincial governments (with local governors at the subprovincial level implementing different policy regimes). This required that all birth control measures should be supervised by provincial governors, allowing them to set locally adjusted “tax rates”, only requiring formal approval from the Provincial Standing Committee of the People’s Congress (Huang et al., 2023). The local adjustment to the policy came as a result of complaints from the population, which resulted in the lowering of fines in some provinces between 1982 and 1984 (Huang et al., 2021).

Regional Variation

To ensure compliance to the policy’s objectives, “social compensation fees” (shehui yangfei) were implemented to discourage fertility and enforce birth quotas (Huang et al., 2023). The penalties were levied on households that exceeded government-mandated birth limits. Their structure and enforcement evolved significantly over time and varied across and within provinces. Certain regions with a predominantly minority population, including Tibet, were generally exempt from fines or granted larger birth quotas (Zhang, 2017). These exemptions generally persisted across the country throughout the period of OCP implementation. Additionally, urban enforcement was stricter in general, with penalties applied even for second births (regardless of the sex of the first child), whereas rural areas (in many cases) permitted exemptions to the policy through additional-child permits (e.g., a second child if the first was a girl).

Penalty structures also diverged in design. Urban areas frequently imposed recurring annual fines until a child reached adolescence. Fines were primarily income-based, tied to local per capita disposable income statistics published by provincial governments. For instance, a common calculation method in Guangdong was a 10 % ‘child tax’, deducted from labor income from each parent each year for a period of 14 years. Some rural areas sometimes used fixed lump-sum amounts to simplify collection. Rural penalties also included non-financial sanctions, such as reductions in land allocations or exclusion from subsidized public services (Ebenstein, 2010). This urban-rural divide created uneven fertility incentives with stricter regulation in cities and more relaxed regulations and weaker enforcement in rural villages (Zhang, 2017).

Temporal Variation

The evolution of fines aggregated at the provincial level is illustrated in Figure 1. The fine levels are calculated as multiples of average local household labor income and are obtained by discounting the total value of the fines across the years of payment, and are constructed by [Ebenstein \(2010\)](#) for the period 1979-2000. Typically, the fines were equivalent to 1 to 2 times local annual household income, but in certain provinces and after the increase in fines in the 1990s, fines could reach up to a present value of 5 to 6 times local annual wage income (e.g Beijing and Xinjiang).

While there has been little to no change in penalties during the early years of the policy until 1989 (except some provinces lowering the fines before 1984), many provinces significantly increased the level of fines in the early 1990s to further curb population growth. Between 1989 and 1992, more than half of the provinces saw a significant increase in the penalty rate, from 0.8 to 3.0 times annual household incomes. This came after former premier Li Peng’s speech about the population “remaining in a race with grain, the outcome of which would affect the survival of the Chinese nation” ([Huang et al., 2023](#)). The increase in fines at the time was most likely the result of a promotion incentive among provincial governors, as [Huang et al. \(2021\)](#) find a strong correlation between the increases in penalty rates and the incidences of government succession. [Li et al. \(2025\)](#) link this to the introduction of the One-Vote Veto system to career incentives, which tied provincial governors’ promotions to meeting population control targets, thereby incentivizing stricter enforcement and higher fines. Fines became increasingly progressive, disproportionately affecting higher-income urban workers. For example, penalties in Beijing reached 6 times annual income, while poorer rural regions retained lower multiples. The 2016 transition to a Two-Child Policy shifted penalties to third births, though wealthier cities such as Shenzhen maintained high fines to deter affluent families from giving birth. In 2021, fines were generally abolished under the Three-Child Policy.

2.3 Family planning before the one-child policy

The One-Child Policy was not the sole contributor to the fertility decline across China. The time period of the policy overlapped with rapid economic growth, and fertility rates across China already saw steep declines before the introduction of the OCP in 1979 ([Zhang, 2017](#)). The “Wan Xi Shao” or “Later, Longer, Fewer” (LLF) campaign already set ambitious national birth targets in the early 1970s, albeit without strict penalties for noncompliance ([Chen and Fang, 2021](#); [Chen and Huang, 2020](#); [Rao and Zhang, 2024](#)). The campaign was launched with first pilots in 1969, and

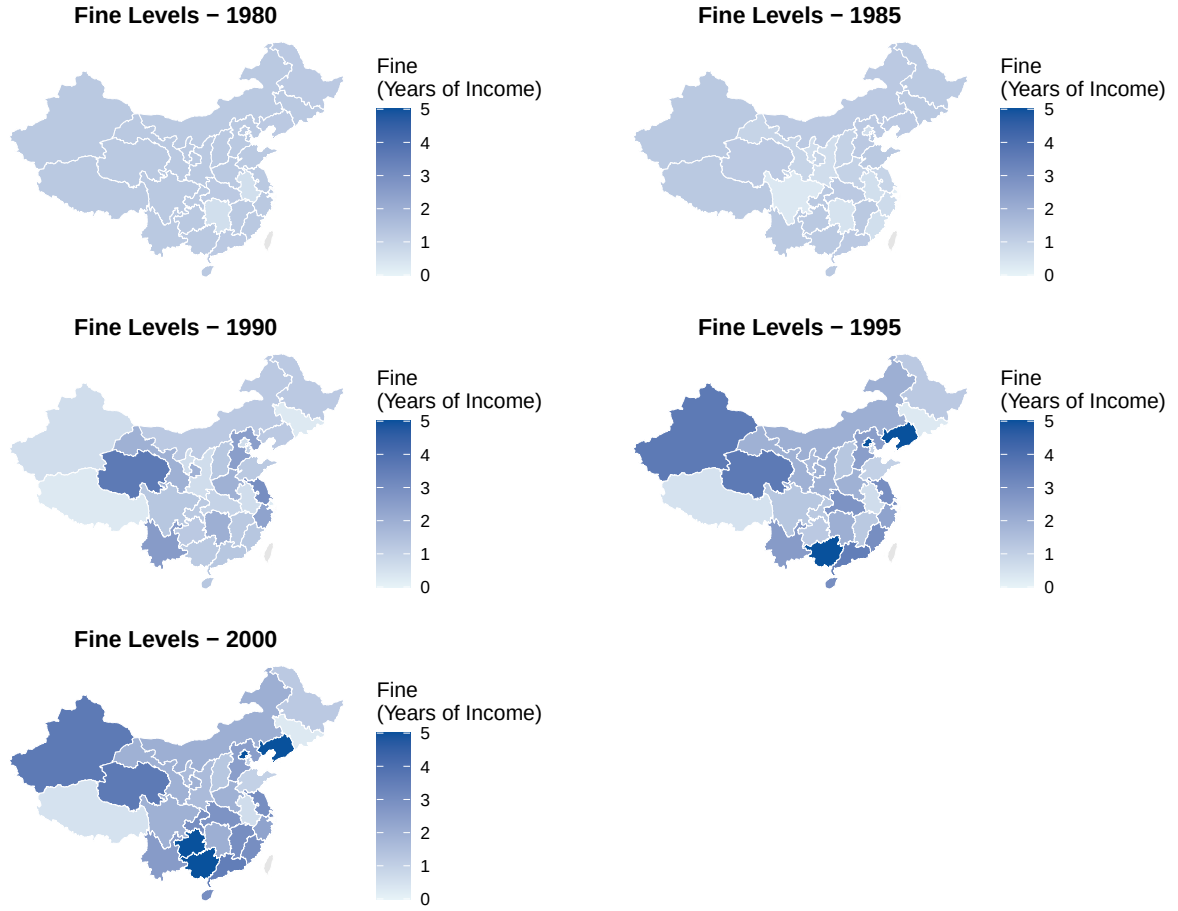


Figure 1: One-Child Policy fine levels by province, 1980-2000 (5-year intervals)

Notes: The figure illustrates the geographic distribution of fine levels (measured in years of income) imposed for violations of the One-Child Policy across Chinese provinces at five-year intervals from 1980 to 2000. Regions shown in gray (Taiwan, Hong Kong, and Macao) have no fine data available. The blue gradient indicates fine intensity, with darker blue representing higher fines.

promoted delayed marriages (minimum age 25 for men, 23 for women in rural areas; 28 and 25 in urban), longer intervals between births, and fewer children overall - “ideally two in urban areas, three in rural”. Later, the campaign would encourage one child, albeit allowing up to two children. The early drop in fertility is visualized in Figure 2. While the average number of children among mothers born between 1930 and 1940 (thus entering childbearing years around 1950 and 1960) was more than four children per woman, this number already started dropping for cohorts in 1940 who completed their childbearing years around 1975.

The implementation of the LLF campaign was primarily overseen by newly established “Family Planning Leading Groups” which operated through workplace units (danwei) and rural communes. These groups emphasized persuasion, propaganda, and peer pressure rather than relying on strict legal coercion. In stark contrast, the One-Child Policy (OCP) instituted after 1979 represented a shift to strict penalties, even resulting in sterilizations and forced abortions in some cases (Zhang,

2017). The OCP essentially transformed recommendations into mandatory quotas with penalties imposed on noncompliant households. In cities, where the danwei system allowed for intense surveillance, compliance was enforced through a system of powerful economic incentives and punishments—including significant fines, loss of employment, denial of social benefits and in some cases even forced abortions or abducting of unauthorized children, that made deviating from the child quota extremely costly for individuals (Baochang et al., 2007; Ebenstein, 2010; McElroy and Yang, 2000).

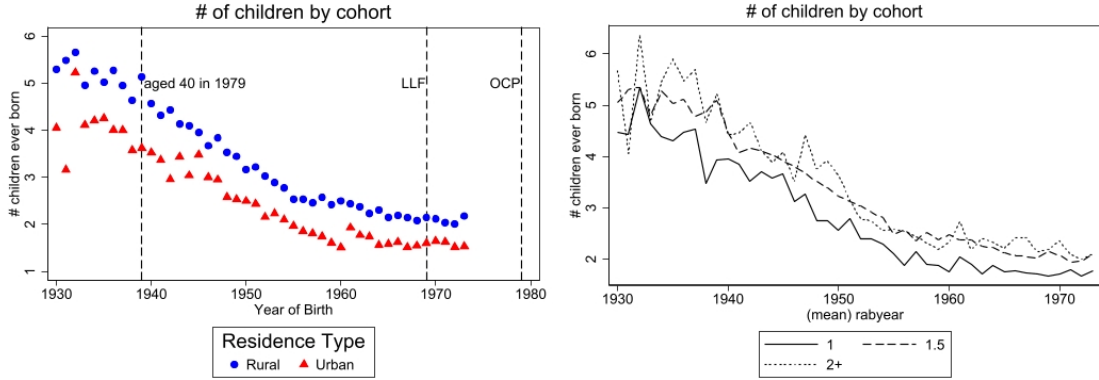


Figure 2: Average number of children by birth cohort, rural/urban areas and policy areas

Notes: The figure illustrates the average number of children born to mothers of different cohorts, by rural areas and One-Child Policy areas. The right plot stratifies the sample on variations of the one-child policy as reported by community heads at the first wave of CHARLS in 2011.

2.4 Policy Exposure by Cohort

Given the policy environment, it is important to identify individuals who were exposed to the One-Child Policy. Empirically, these are the potential women who made their decision to proceed to a second child *after* the introduction of strict fertility limits in 1979.

Cohorts born before 1933 were not exposed any family planning (neither LLF or OCP), as they generally concluded their childbearing years prior to their launches in 1969 and 1979. Cohorts born between 1933 and 1942 were primarily exposed to LLF incentives during the later stages of their reproductive lives. The One-Child Policy, introduced in 1979, subsequently imposed strict constraints on women born after 1942: those born in the 1950s encountered the policy later, while cohorts born in the 1960s and 1970s were the first to be subject to the OCP throughout their entire reproductive window.

Women who were 37 or older in 1979 (these are cohorts born before 1942) can be considered

fully untreated cohorts of women with respect to the One-Child Policy. Of course many women who wanted to have a second child would already have done so before the age of 37, but after this age, virtually all individuals among cohorts born before 1942 would already have had a second child (Li and Shi (2025) note that around 97% of women born in the 1935-1945 cohorts already had a second child by age 37). In my sample, the mean age at last birth is 37 in rural areas and 33 in urban areas for cohorts born before 1930. The mean age at second birth was around 26 (see Appendix Table B.3 for details).⁴

3 Data

3.1 The China Health and Retirement Longitudinal Study (CHARLS) (2011-2018)

The data come from the China Health and Retirement Longitudinal Study (CHARLS). CHARLS is the first Chinese nationally representative survey of the elderly population aged 45 and above, and collects information on demographics, employment, retirement, pensions, family transfers, physical and mental health, as well as biomarker information for approximately 20,000 individuals in 10,000 households. CHARLS is designed along the same lines of the Health and Retirement Study in the United States (HRS), and follows individuals in mostly biennial intervals.⁵ Currently, the waves from 2011, 2013, 2015, 2018, and 2020-23 are publicly available.

CHARLS stratifies by region, urban/rural districts and GDP per capita. In total, 150 counties from all Chinese provinces except Tibet were randomly selected using probability-proportional-to-size (PPS) sampling. From each county, three primary sampling units (PSUs), or ‘communities’, consisting of urban neighborhoods (shequ) or villages (cun) were chosen using probabilities proportional to their population. Within each community, households from randomly selected housing units were interviewed if they had at least one member aged 45 or older. Within each household, CHARLS interviewed a main respondent and their spouse, if available. At baseline, this resulted in a sample of 17,708 individuals, living in 10,257 households, located in 450 resident communities (300 villages and 150 urban neighborhoods) from 150 counties or urban districts. Respondents who were not present at the first wave were still contacted in the follow-up waves. The dropout rate is

⁴The high age at birth among older cohorts may come at a surprise. I focus on live births, as the number of children that matters to parents when they plan for old age insurance likely is live births. This is consistent with the decision made in Rossi and Rouanet (2016). Live births and children ever born may differ if infant mortality is high - which was the case among older cohorts who gave birth during the Great Famine.

⁵Other equivalent sister surveys are the Aging and Retirement in Europe (SHARE), the English Longitudinal Survey of Aging (ELSA), the Japanese Study of Aging and Retirement (JSTAR), and the Longitudinal Aging Survey of India (LASI). To obtain the CHARLS data, researchers need to request access from <https://charls.charlsdata.com>.

approximately 15%, which is comparable to that of retirement surveys in other countries.⁶

3.2 Fines for unauthorized births (1979-2000)

I obtain fines imposed during the One-Child Policy from [Ebenstein \(2010\)](#). The dataset contains a province-year panel of fines for unauthorized births in multiples of household income, as well as variation in policy regimes from 1979 to 2000. I merge this data to CHARLS data using province identifiers after dropping Tibet, Hainan and Ningxia, because the latter provinces are not sampled within CHARLS.

3.3 Cognition

Defining and measuring cognitive impairment in population-based surveys generally presents challenges, as individuals with severe impairment may be institutionalized or unable to participate in surveys. In my CHARLS sample, nursing home residence is extremely low (0.5%) and most individuals are interviewed at home. Unlike the Health and Retirement Study (HRS), which includes a validated dementia subsample, the China Health and Retirement Longitudinal Study (CHARLS) lacks a directly comparable clinical validation group.⁷ I therefore construct a composite measure of cognitive function based on three informative components that are available consistently across waves: immediate and delayed word recall (number of words correctly recalled from a 10-word list, ranging from 0 to 20), serial 7s subtraction test (number of correct subtractions from 100, ranging from 0 to 5), and temporal orientation (correctly naming the date, day of the week, month, and year, ranging from 0 to 4).⁸

I proceed by constructing two measures of cognition. First, I compute a summary score by summing the correct responses across all three scales (ranging from 0 to 29). Similar to the approach taken for HRS respondents by [Langa et al. \(2023\)](#), I classify respondent-time observations into three clinical categories: probable dementia (score 0–6), cognitive impairment not demented or CIND (score 7–11), and normal cognition (score 12–29).⁹ Second, to obtain a continuous measure that is

⁶In the second, third and fourth waves, respectively, 86%, 82% and 77% of the original participants who responded to the first wave were present.

⁷In 2018, China administered the HCAP module which included a MMSE (Mini-Mental State Examination) with detailed cognition measures. However, as the CHARLS data team report in a 2023 data alert, this MMSE was validated only among people who can read and write, have no vision impairments, and spoke Mandarin (despite being widely taught, not all Chinese elderly can speak Mandarin). Therefore, they note that the data likely contains biased measurements of cognition.

⁸I do not use the available scales on self-reported cognition and the ability to copy an assigned picture because these are not commonly used in research comparable to this paper (an overview of the available cognition scales in CHARLS is reported for each age group of the complete raw sample in Appendix Table C.1).

⁹For the latter category, [Langa et al. \(2023\)](#) use the range 12-27 on the total score on immediate and delayed word recall, serial 7s subtraction test and backward counting (the latter test is unavailable in CHARLS).

interpretable and comparable across waves, I construct a standardized cognitive score. I standardize the total raw score (0–29) to have a mean of 0 and a standard deviation of 1 across the pooled sample. The distribution of cognition across the sample is shown in Figure 3: the distribution is moderately skewed to the left, indicating a higher density of individuals with good cognition (skewness of -0.33). I also visualize the evolution of cognition by age and time in Appendix Figure C.2. Age-adjusted time trends by gender are available in Appendix Figure C.3.

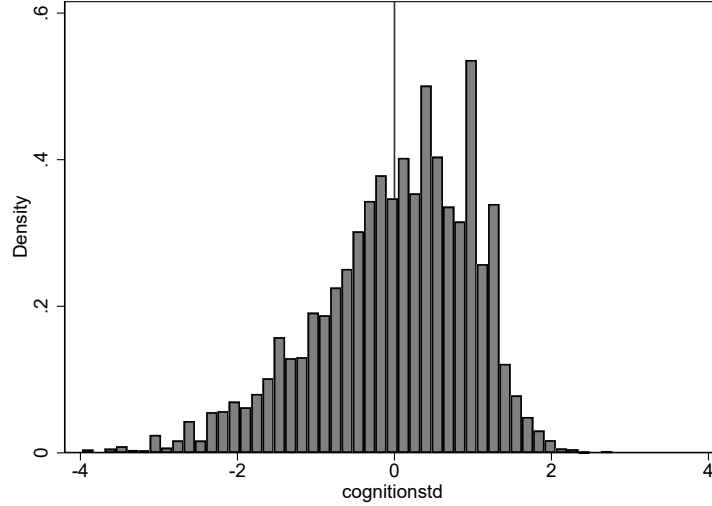


Figure 3: Standardized Cognition

Notes: Data come from the China Health and Retirement Longitudinal Study (CHARLS).

3.4 Additional Outcomes

In addition to cognitive health, I examine secondary economic outcomes to explore potential mechanisms. Labor supply is measured at both the extensive margin (whether the individual works) and intensive margins (weeks worked in the past year and hours worked in the past week). Inter-generational transfers are measured as whether the individual receives any financial support from children and grandchildren, whether a child coresides in the same city or county, and the total amount of financial support received in the past year. To reduce the influence of extreme values, continuous variables including weeks worked, hours worked, and transfer amounts are winsorized at the 99th percentile.

3.5 Stopping Probability and Birth Spacing

The share of CHARLS respondents living in predominantly 1, 1.5 and 2-child policy provinces are comparable to those from census data reported in [Ebenstein \(2010\)](#).¹⁰ Among the 450 sampled communities in CHARLS, 48 percent of community heads in 2011 reported to be implementing a 1.5 child policy (comprising 53% of individuals in the sample), 41 percent a one-child policy (35% of individuals), and 11 percent (11% of individuals) a 2- or 3- child policy.¹¹

Table 1 and Appendix Tables B.1 and B.2 show the sample average probability of having an additional child conditional on the sex of children at lower parities, stratified by birth cohort. These patterns are also visualized in Figure 4 for ease of interpretation. The data reveal increasing son preference among cohorts impacted by the One-Child Policy, particularly towards the second child. While mothers born between 1930 and 1950 show small or no son preference in their decision to have a second child, this gap widens significantly for younger cohorts. For mothers born in the 1970s living in rural areas, the probability of having a second child is substantially higher if the firstborn is female. This is most pronounced in 1.5-child policy areas, where the continuation probability is 91% for those with a firstborn daughter compared to 52% for those with a firstborn son—a 39 percentage point gap (Table 1). In urban areas, the gap is also present but smaller, around 16–17 percentage points (Appendix Table B.1). This distinct pattern indicates that after the introduction of strict fertility limits and associated fines, the decision to proceed to a second child became strongly dependent on the sex of the firstborn for these cohorts. Similar patterns of son preference drive the decision to proceed to a third child, with the highest continuation probabilities observed among families with two daughters (Appendix Table B.2).^{12 13}

¹⁰[Ebenstein \(2010\)](#) reports a current population share of 54% living in 1.5 child areas, 35% in one-child areas and 10% in two child areas around 2010.

¹¹Appendix Tables D.1 and D.2 contain a full list separated by rural/urban classification status of the community. Appendix Table D.5 contains a detailed list of policy type heterogeneity across provinces. Appendix Table S.2 contains a summary list of which provinces predominantly had which variation of the One-Child Policy in place (as reported in [Ebenstein \(2010\)](#)). A graphical visualization of this is available in Appendix Figure S.2.

¹²It is important to note that in strict one-child areas the reported firstborn child's sex may be inaccurate because of sex-selective abortion and infanticide in these areas during the 1980s and 1990s, due to wider access to ultrasound technology. I thus focus primarily on 1.5-child areas.

¹³Appendix Figure B.3 also shows fertility continuation probabilities by firstborn gender.

Table 1: Probability of second child in Rural areas by policy area

Policy Area / Firstborn Sex	Birth Cohort				
	1930	1940	1950	1960	1970
1-child					
Female	0.99	0.98	0.91	0.88	0.78
Male	0.95	0.98	0.92	0.74	0.70
<i>Difference (F-M)</i>	0.03	0.00	-0.01	0.14	0.08
1.5-child					
Female	0.99	0.96	0.96	0.92	0.91
Male	0.98	0.99	0.89	0.72	0.52
<i>Difference (F-M)</i>	0.01	-0.03	0.07	0.19	0.39
2+-child					
Female	1.00	0.98	0.93	0.91	0.86
Male	1.00	0.99	0.91	0.86	0.84
<i>Difference (F-M)</i>	0.00	-0.01	0.01	0.06	0.01

Notes: Table shows the probability of having a second child by firstborn sex for rural mothers across all OCP policy areas. For each policy area, the Difference row shows son preference (Female - Male). Positive values indicate higher continuation probability when the firstborn is female. The strongest son preference effects are observed in 1.5-child policy areas among 1970s cohorts (39 percentage points).

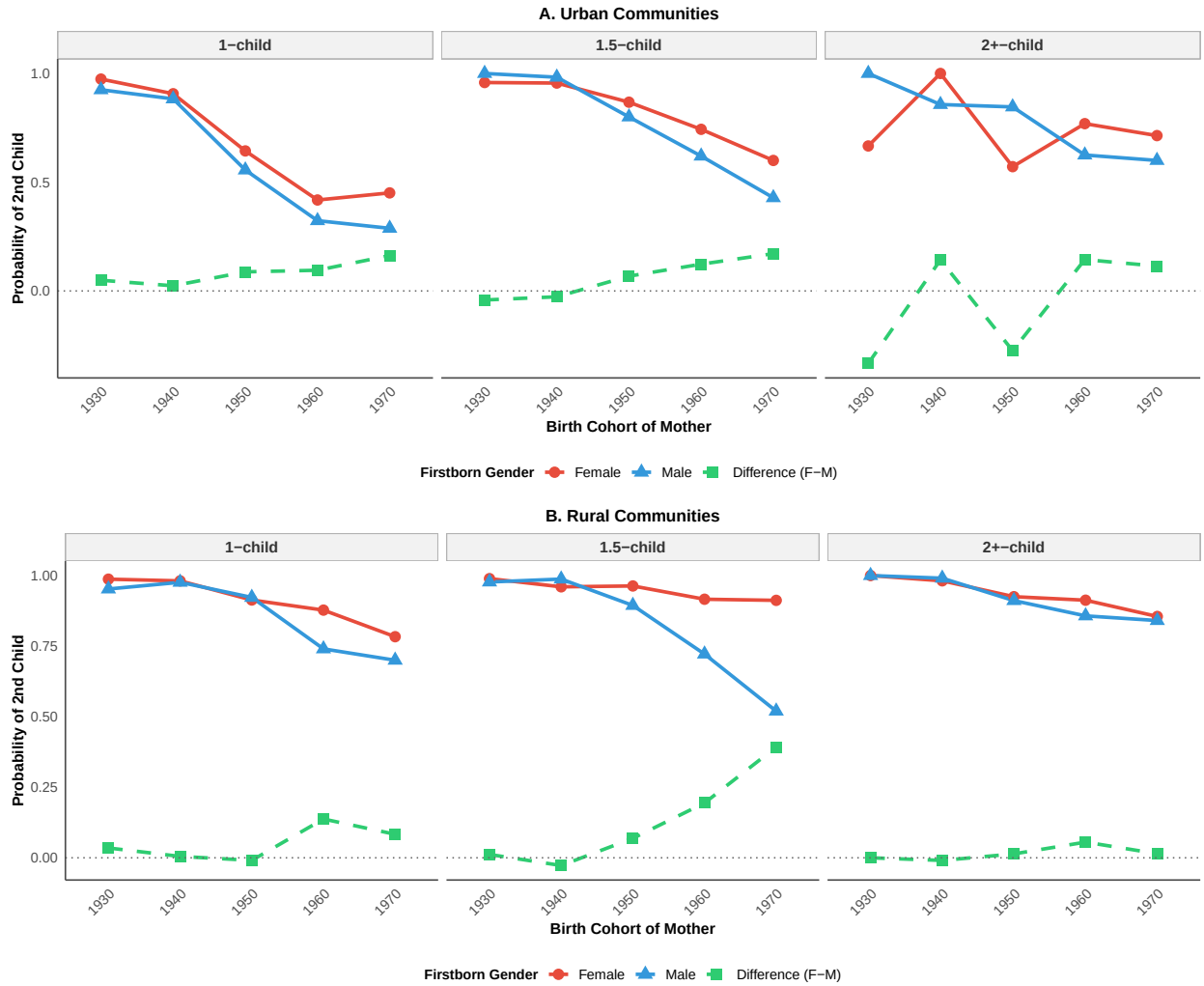


Figure 4: Fertility continuation patterns and son preference by urban and rural communities and policy area

Notes: Panel A shows patterns in urban communities, Panel B in rural communities. Each panel displays the probability of having a second child for mothers with female (red) vs. male (blue) firstborns across birth cohorts and OCP policy areas. The green dashed line shows the son preference gap (Female - Male difference). Rural areas demonstrate substantially stronger son preference effects than urban areas, particularly in 1.5-child policy areas among 1970s cohorts where the rural gap reaches 39 percentage points compared to 17 percentage points in urban areas. Note that few people live in 2+ child urban areas, which explains the imprecise estimates for this group.

3.6 Sample Selection

The unit of observation is the elderly parent at risk of cognitive decline. Because child-mother combinations are generally safer to match than child-father (due to potential separation of parents resulting in the mother taking care of children, and due to fathers in rural China being more likely than mothers to engage in work in remote cities), I focus the main analysis on mothers. I restrict the analytic sample to mothers born between 1930 and 1973 who have at least one child (cohorts born in 1973 were 45 in 2018). I also drop mothers who had their first birth after 2001 because I do not have information on the financial penalties after 2000, leaving a final sample of 35,637 total mother-time observations.¹⁴ A detailed overview of the sample selection process is available in Appendix Tables A.1 and A.2, with a Lexis diagram in Appendix Figure A.1

In the analytic sample, to have a clean sample I keep only mothers from 1.5 child areas and thus exclude mothers from strict one-child areas, because the latter already faced restrictions at the time of the first birth. I make this decision because facing restrictions at the first parity would affect the propensity of parents to abort the child, given increased access to ultrasound in the 1980s, which would invalidate the key assumption of random sex assignment of the first child (Chen et al., 2013; Huang et al., 2024). The sample is restricted to observations from individuals aged 45 and above.

3.7 Summary Statistics

I show summary statistics of demographic information for mothers by policy area in Table 2 for cohorts born between 1950 to 1974. Mothers born in the 1950s completed their final fertility years after the introduction of the OCP in 1979. Younger cohorts completed most of their fertility decisions under active policy enforcement.

It is evident that among younger cohorts born after 1950 in Table 2, the average number of living children is notably higher in regions with a more relaxed policy rule in place. Mothers in 1.5-child policy areas have on average 2.40 children, compared to 2.00 children in stricter one-child areas (in 1.5 child areas, 66% have at least two children compared to only 46% in 1-child areas). Clearly, more relaxed-policy areas are predominantly low-educated, are more likely to be rural, and

¹⁴Note that this restriction is not binding, since CHARLS respondents younger than 45 in the sample tend to be the younger spouses of older husbands and are not representative of the population. Respondents between 2011 to 2018 had in the vast majority of cases their first child prior to 2002 (the youngest representative cohorts aged 45 in 2018 (born in 1973) were 29 in 2001, much older than both the mean and median age at second live birth, which was about 26 years as shown in Appendix Table B.3). Appendix Figure B.4 plots the distribution of age at first birth across cohorts and Appendix Figure B.2 visualizes the spacing of births by birth cohort of the mother. The age at first marriage and at first birth is plotted against the year of occurrence in Appendix Figure B.5. I do not observe the age at menarche, which has previously been used as an instrument for fertility in many developing countries (Field and Ambrus, 2008).

are comparatively poorer.

Table 2: Mother characteristics by one-child policy area

	Policy Han (2011)		
	1 11,964 (36.0%)	1.5 17,606 (52.9%)	2+ 3,706 (11.1%)
age	55.08 (6.05)	54.64 (5.94)	54.36 (5.97)
male	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cognition (Z-score)	0.27 (1.01)	-0.02 (0.99)	-0.30 (1.05)
Cognition Total	15.10 (5.40)	13.56 (5.34)	12.03 (5.63)
Cognitive Impairment (cutoff)	0.24 (0.43)	0.33 (0.47)	0.46 (0.50)
Cognitive Impairment (Bottom 20%)	0.16 (0.37)	0.22 (0.42)	0.33 (0.47)
working	0.58 (0.49)	0.71 (0.45)	0.79 (0.41)
# of living children	2.00 (1.02)	2.40 (1.06)	2.37 (1.01)
has at least 2 children	0.46 (0.50)	0.66 (0.47)	0.69 (0.46)
number of living daughters	0.96 (0.90)	1.12 (0.96)	1.16 (0.92)
number of living sons	1.03 (0.79)	1.27 (0.80)	1.21 (0.84)
any child in same city/county	0.74 (0.44)	0.76 (0.42)	0.82 (0.38)
any in-person contact	0.75 (0.43)	0.73 (0.44)	0.79 (0.41)
net asset quartile	2.86 (1.11)	2.60 (1.04)	2.46 (1.04)
education level			
education: primary	9,246 (77.3%)	16,538 (94.0%)	3,544 (95.7%)
education: college	2,715 (22.7%)	1,060 (6.0%)	159 (4.3%)
cohort group (10-year)			
1950	5,600 (46.8%)	7,740 (44.0%)	1,532 (41.3%)
1960	5,301 (44.3%)	8,316 (47.2%)	1,778 (48.0%)
1970	1,063 (8.9%)	1,550 (8.8%)	396 (10.7%)

Notes: The table shows means with standard deviations in parentheses for continuous variables, and counts with percentages for categorical variables. Sample: mothers born 1950-1979.

Next, using data from 1.5-child areas, I also describe mothers' characteristics by gender of the firstborn child in Table 3, using only mothers born between 1960-1969 for illustration. Household composition and fertility show clear differences: mothers with a female firstborn have more children on average (through more daughters and fewer sons). These patterns are consistent with son-preference-driven fertility behavior. They also report slightly higher rates of co-residence and weekly in-person contact with children, and higher labor force participation, while net-asset quartile and education levels are broadly similar across groups. Individuals do not seem to differ in their performance on cognitive tests based on their firstborn gender. However, as Appendix Figure C.4 illustrates, there is a clear relationship between the number of children and the cognition score across birth cohorts, with more children being associated with worse performance across ages (except for 1930 cohorts where mothers of two children have better cognition than one or three or more children - a similar pattern observed in the literature on the EU and US for later cohorts (Bonsang and Skirbekk, 2022)).

Table 3: Mother characteristics by sex of firstborn

	1 hachgendr		Test
	0	1 Male	
	2,117 (52.2%)	1,938 (47.8%)	
age	50.28 (3.22)	50.26 (3.14)	0.838
male	0.00 (0.00)	0.00 (0.00)	.
Cognition (Z-score)	0.21 (0.92)	0.22 (0.93)	0.797
Cognition Total	14.77 (4.92)	14.81 (5.01)	0.797
Cognitive Impairment (cutoff)	0.24 (0.43)	0.23 (0.42)	0.673
Cognitive Impairment (Bottom 20%)	0.15 (0.36)	0.15 (0.35)	0.768
working	0.79 (0.41)	0.74 (0.44)	0.001
# of living children	2.40 (0.92)	1.92 (0.82)	<0.001
has at least 2 children	0.89 (0.31)	0.69 (0.46)	<0.001
number of living daughters	1.60 (0.77)	0.45 (0.61)	<0.001
number of living sons	0.78 (0.67)	1.47 (0.61)	<0.001
any child in same city/county	0.78 (0.41)	0.73 (0.44)	<0.001
any in-person contact	0.76 (0.42)	0.72 (0.45)	0.002
net asset quartile	2.76 (0.96)	2.73 (1.01)	0.349
rural hukou	0.93 (0.25)	0.90 (0.31)	<0.001
education level			
education: primary	1,902 (89.8%)	1,764 (91.0%)	0.203
education: college	215 (10.2%)	174 (9.0%)	
cohort group (10-year)			
1960	2,117 (100.0%)	1,938 (100.0%)	.

Notes: The table shows means with standard deviations in parentheses. The rightmost column shows p -values from t -tests comparing female and male firstborn groups. Sample: mothers born in 1960s living in 1.5-child policy areas.

4 Empirical Approach

Estimating a credible causal effect of fertility on cognition is difficult because the decision to have more children is inherently correlated with unobserved individual and household characteristics, such as preferences for children, child investments, or innate cognition, which would bias regression estimates if these issues are not properly accounted for. The policy setting during the one-child policy generates plausible exogenous variation in family size, because households making fertility decisions incorporate the cost of financial penalties when optimizing fertility under fertility constraints. When the firstborn child does not have the preferred gender in the presence of son preference, parents proceed to have an additional child hoping that the desired fertility outcome (at least one son) would be achieved at a higher parity, even if this implies having to incur a fine.

4.1 Fines as price mechanism of unauthorized births

Following a similar conceptualization as [García \(2024\)](#), I characterize the social compensation fee as the price of a marginal child at the second parity for a birth that exceeds the legal quota (in addition to household-specific expected childrearing costs). Although I do not estimate the structural model, I formally present the theoretical framework for ease of interpretation.

Let q denote the legal quota of children (e.g., $q = 1$ or $q = 1.5$) and ω_i the number of children for household or individual i .¹⁵ Denote by $f_{p,a}$ the province-specific fine faced by a woman at age a in province p , measured as a fraction of annual household income. A household chooses ω to maximize lifetime utility net of childbearing costs:

$$\max_{\omega \in \mathbb{Z}_+} U_i(\omega; Z_i, X_i) - C_i(\omega; X_i) - \mathbf{1}\{\omega > q_p\} f_{p,a},$$

where $U_i(\omega; Z_i, X_i)$ denotes the household's utility when it has ω children and $C_i(\cdot)$ captures non-policy related childbearing costs; I characterize the marginal benefit only at the second parity as $\Delta U_{i,2} = U_i(2) - U_i(1)$, which may depend on firstborn sex Z_i , fines $f_{p,a}$, and covariates X_i .

The decision to have a second child is therefore discrete and governed by the threshold condition

$$\Delta U_{i,2} - \Delta C_{i,2} > f_{p,a},$$

where $\Delta C_{i,2} = C_i(2) - C_i(1)$. Son preference implies $\mathbb{E}[\Delta U_{i,2} \mid Z_i = 1] > \mathbb{E}[\Delta U_{i,2} \mid Z_i = 0]$, so a firstborn daughter raises the left-hand side and increases the probability of a second birth; conversely, a larger fine $f_{p,a}$ lowers that probability. Empirically, this yields the testable implications used in the first stage: a positive effect of Z_i on the likelihood of a higher parity and a negative effect of $f_{p,a}$ on that likelihood, justifying the role of fines as a price instrument for unauthorized higher-order births.¹⁶

I estimate the causal effect of fertility on late-life cognition within a potential outcomes framework. Let the treatment be defined as $\mathcal{F}_{ica} \in \{0, 1\}$, where $\mathcal{F}_{ica} = 1$ denotes a parity of two or more children and $\mathcal{F}_{ica} = 0$ denotes one child (I exclude individuals with no children).¹⁷ Since I

¹⁵For simplicity, I use household and individual interchangeably, acknowledging that intrahousehold bargaining may affect the decision-making process.

¹⁶I focus on 1.5 child areas because the penalty structure in these areas incentivize parents of a daughter to have a second child in hopes that it would be a boy. The sex of the firstborn child is likely to be close to random in these areas because sex-selective abortion at the first parity was not widespread, whereas it was prevalent in strict one-child areas, in particular after the 1990s ([Chen et al., 2013](#); [Ebenstein, 2010](#); [Huang et al., 2024](#)).

¹⁷I define the treatment as crossing the 1 to 2 parity threshold (≥ 2 children) rather than having exactly two children, because the instruments operate at this margin. Restricting to exactly two children would condition on a

only observe one potential outcome for every individual i , $Y_{ict} = \mathcal{F}_{ica}Y_{ict}(1) + (1 - \mathcal{F}_{ica})Y_{ict}(0)$. A naive comparison of means after controlling for covariates, $\mathbb{E}[Y_{ict}|\mathcal{F}_{ica} = 1, X_i] - \mathbb{E}[Y_{ict}|\mathcal{F}_{ica} = 0, X_i]$, is confounded by unobserved heterogeneity in preferences, health, and socioeconomic factors that jointly determine fertility decisions and cognitive aging. To address this endogeneity, I specify a linear model for the observed cognitive outcome of individual i in county c at wave t , whereby $\mathcal{F}_{ica}$ is an endogenous regressor, because the decision to proceed to have a second child depends on the (dis)incentives an individual was exposed to during childbearing age. The second stage is defined as:

$$Y_{ict} = \beta_0 + \tau \hat{\mathcal{F}}_{ica} + \mathbf{X}'_{it}\boldsymbol{\beta} + \theta_t + \theta_c + \theta_c \times \text{rabyear}_i + \epsilon_{ict} \quad (1)$$

I identify the causal parameter of interest τ in Equation (1) by means of Two-Stage Least Squares (2SLS). The vector $\mathbf{X}_{it}$ contains control variables that include a linear and quadratic term in age, and education. I absorb aggregate shocks common to all individuals in a given county with θ_c , and time-specific differences with θ_t . I control for county-specific cohort trends in the outcome linearly with $\theta_c \times \text{rabyear}_i$.¹⁸ The term $\epsilon_{ict} = \alpha_i + \eta_{ict}$ contains the idiosyncratic error composed of a random error and individual unobserved heterogeneity α_i (e.g., genetic endowment), which would bias the estimated treatment effect in ordinary least squares regression without instrumenting the endogenous variable, $\mathcal{F}_{ica}$, with a valid instrument. The term $\hat{\mathcal{F}}_{ica}$ is the projection of $\mathcal{F}_{ica}$ on the two instruments presented below. I thus define the first-stage regression as:

$$\mathcal{F}_{ica} = \gamma_0 + \gamma_1 Z_i + \gamma_2 \left[(1 - Z_i) \cdot \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k} \right] + \mathbf{X}'_{it}\boldsymbol{\gamma} + \delta_t + \delta_c + \delta_c \times \text{rabyear}_i + \epsilon_{ict} \quad (2)$$

where the first instrument Z_i is a binary indicator for a firstborn daughter ($Z_i = 1$), inducing exogenous variation in $\mathcal{F}_{ica}$ through son preference. The term $\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}$ refers to the average fine level (in terms of discounted household income) that woman i faced in the two years ($n = 2$) after the first successful pregnancy (subscript a defines the age at first birth). For example, if a woman first gave birth to a boy at age $a = 23$, then the average fine present in the province between age 23 and 24 is used, since a higher fine would discourage proceeding to a second birth (which would follow approximately 2 years after the first birth, according to birth spacing patterns in relaxed 2-child policy areas and in pre-treatment cohorts of the sample). If a woman gave birth to a girl at the first parity, no fine would be incurred for a second child, as the household would be exempt from the fine making the interaction term zero by construction (Scharping, 2013). The

post treatment outcome (the decision not to have a third child), introducing selection bias.

¹⁸In a robustness check I also add a quadratic trend, but this does not change results substantially.

second instrument, $(1 - Z_i) \cdot \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}$, explicitly captures that fines only deter second births among parents whose firstborn was male. This specification avoids including three instruments (Z_i , $\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}$, $Z_i \times \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}$) when only two sources of variation exist: son preference and fines conditional on having a male firstborn.¹⁹ Since household assets may affect the ability to pay fines for above-quota births, I include the quartile of mean household assets for the individual in the period 2011 to 2018 as a control in robustness analysis.

My main specification restricts the sample to mothers born 1950 or later, focusing on cohorts whose entire childbearing period fell within the OCP era and avoiding survivors of the Great Famine whose cognition may be confounded by early-life nutritional deprivation. Since both instruments Z_i and $\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}$, as well as the endogenous regressor $\mathcal{F}_{ica}$, are time-invariant within the individual, individual fixed effects cannot be used: they would absorb the treatment and instruments entirely, leaving τ unidentified. All causal identification therefore comes from variation between individuals. The panel structure of CHARLS, where cognition is measured repeatedly across waves, provides the advantage of estimating a pooled IV model, which I use as my primary specification. I use all available person-wave observations, absorb wave and county fixed effects, and cluster standard errors at the individual level. Since both instruments are time-invariant, individual fixed effects would absorb them and identification would be lost. I therefore exploit the panel dimension through a pooled IV approach, using all available person-wave observations and clustering standard errors at the individual level. The pooled IV estimator exploits between-person variation in the instruments while retaining the full panel, improving precision relative to a single cross-section.

In addition to the pooled IV approach, I also include alternative estimators. These include a (i) 'between' individuals IV that averages cognition $\bar{Y}_{ic} = T_i^{-1} \sum_t Y_{ict}$ within individuals across time, yielding one observation per person, (ii) a first-observation estimator that keeps only each individual's earliest wave with a non-missing cognition score and runs cross-sectional 2SLS, (iii) a Mundlak correlated random effects IV that augments the pooled specification with individual-level means of the time-varying covariates, approximating individual fixed effects for time-varying confounders while preserving identification from the time-invariant instruments.

4.2 Identification Concerns

For the chosen instruments to be valid, four conditions need to be satisfied. First, relevance requires that the instrument be a strong predictor of family size. From the evidence reported in descriptive

¹⁹An alternative specification that includes all three terms is available in Appendix Table E.1. The alternative specification confirms that the negative fine effect is indeed driven by firstborn boys (-0.147 vs -0.035 for girls, 76% weaker)

tables above and later supported in the regression output below, this assumption is clearly satisfied. Because of the strong historical preference for sons in China, parents with a firstborn girl were significantly more likely to have a second child hoping for a boy, even if it meant incurring a penalty for doing so. Additionally, higher penalties plausibly disincentivize higher fertility, as it would be hard to justify that parents would opt for larger family size if and only if penalties were higher. Both of these basic facts align with earlier results from census (Ebenstein, 2010) and other survey data (McElroy and Yang, 2000). In low-fertility contexts and in particular in the presence of fertility restrictions, such as the Chinese cohorts under consideration in this paper, the gender of the child at first parity (rather than higher-order child sex compositions) is already sufficient to induce substantially different fertility decisions among parents (Angrist et al., 2010).

Second, exogeneity requires the instruments to be unrelated to unobserved individual factors affecting cognition. The sex of the firstborn child is effectively random in the absence of sex-selective abortion, making it uncorrelated with individual characteristics. Fines for unauthorized births can be considered exogenous if they are determined by local policy rather than individual traits. A potential concern is that regions with higher demand for children might have imposed stricter fines, resulting in fines being correlated with fertility preferences. Huang et al. (2021) and Huang et al. (2023) find no significant correlation between fine implementation dates and provincial macroeconomic indicators (e.g., GDP, education) or fertility trends, suggesting that fines were not systematically endogenous to local conditions.²⁰

Even if local officials balanced national objectives with local demand, this paper estimates a local average treatment effect at the second parity by comparing parents within the same counties who differ only in the sex of their firstborn. By controlling for county-specific (sub-prefectural) factors, the variation in fines across time thus captures the disincentive for additional births, independent of individual-specific fertility preferences.

The monotonicity assumption (which is not required for consistency of the estimator but is required for LATE interpretation) requires that the instrument must not create defiers. In the present context, this means there should be no parents who would have a second child only if their firstborn was a boy (making them ineligible for a second birth under the 1.5 or 2 policy norm), but would not have a second child if their firstborn was a girl (making them eligible). Similarly, no parents should want to have an additional child in the presence of prohibitively high fines but would not want to do so in absence of the latter. It is thus reasonable to assume all parents either comply with the incentive created by the instrument (have a second child if the first is a girl and

²⁰Huang et al. (2023) use the China Compendium of Statistics (CCS) for macroeconomic indicators prior to 1992.

finer are low) or otherwise never would. This ensures the Local Average Treatment Effect (LATE) is estimated for the group of compliers, i.e. those whose fertility decisions were actually influenced by the instrument.

Finally, the exclusion restriction requires that the instrument influences late-life cognition only through its effect on the number of children. I argue that in principle the sex of the first child and the specific local fine rate should not directly affect a parent's cognitive health decades later, other than by changing fertility decisions. While the sex of the firstborn child affects the household decision of incurring fines for an additional birth, it also may change other circumstances, such as the age at which the elderly parent becomes a grandparent (because parents of daughters on average transition to grandparenthood earlier than parents of sons (Eibich and Zai, 2024; Rupert and Zanella, 2018)), or the fact that older sisters may be more likely than younger brothers to give care to the aging parent than older brothers. However, as I argue in this paper, both of these factors still run through family size in the present context, as child preferences at the time in absence of restrictions were consistently more than two children. To nevertheless address this concern, in robustness analyses I also control for the firstborn's gender in the outcome equation, similar to (Bonsang and Skirbekk, 2022). Another concern related to the exclusion restriction is that parents of children may make different housing choices based on the overall sex composition of children. For example, if parents with one son and one daughter want to keep a separate bedroom for each, parents of mixed-sex children may choose to purchase more spacious apartments (after the birth of children). This in turn may influence elderly cognition - for instance by relocating to residential areas with community-based activities, such as nearby parks to exercise, which are popular among the Chinese elderly. However, since I investigate a primarily rural and poor context, these concerns are less serious.

Relating these assumptions to the above equation, the conditions can be formally defined as: $\text{Cov}(Z_i, \mathcal{F}_{ica}) \neq 0$ and $\text{Cov}(\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}, \mathcal{F}_{ica}) \neq 0$ for relevance, $\mathbb{E}[\alpha_i + \eta_{ict} | Z_i] = 0$ and $\mathbb{E}[\alpha_i + \eta_{ict} | \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}] = 0$ for exogeneity, $\mathcal{F}_{ica}(Z = 1) \geq \mathcal{F}_{ica}(Z = 0)$ and $\mathcal{F}_{ica}(\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k} + \epsilon) \leq \mathcal{F}_{ica}(\frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}) \quad \forall i$ for monotonicity, and finally $Y_{ict} = f(\mathcal{F}_{ica}, X_{it})$ and not $Y_{ict} = f(\mathcal{F}_{ica}, X_{it}, Z_i, \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k})$ for the exclusion restriction. All the conditions assume SUTVA and sufficient support. Under the conditions outlined above, τ recovers the local average treatment effect (LATE) of having a second birth on cognition among compliers, i.e. those whose fertility decision was altered by the instrument.

Similar to previous work (Ebenstein, 2010; García, 2024; Huang et al., 2023; Xiao, 2024), my approach only captures one means of punishment and does not have information about nonfinancial

penalties, such as restrictions on access to public services or reductions in land allocation, or social stigmatization (although these might be correlated with the financial penalties). As long as these nonfinancial penalties do not systematically relate to the outcome (which is cognitive health), this would not be a central concern. As [Ebenstein \(2010\)](#) notes in footnote 18, because such nonfinancial penalties were imposed informally and without a clear structure, they would be essentially random in the data.

5 Results

5.1 Main Results

Because of the institutional setting and stricter birth quotas in one-child policy areas, I focus the main analysis on 214 out of the 450 communities where the community head in 2011 reported to be implementing a 1.5-child policy. Table 4 reports first-stage and second-stage estimates across four estimators: a between-individuals IV (Column 1), the pooled IV (Column 2), a correlated random effects specification (Column 3), and a first-observation cross-section (Column 4). All specifications include county fixed effects; Columns 2 and 3 additionally include wave fixed effects. County-specific linear cohort trends are absorbed in all columns.

Both instruments are individually significant at the 1% level in every specification. In the preferred pooled specification (Column 2), mothers with a firstborn daughter have a 3.9 percentage point higher probability of proceeding to a second birth, while a one-unit increase in the fine rate among mothers of firstborn sons reduces this probability by 12.1 percentage points. The interaction term captures the fact that fines deter second births when the firstborn is a son, since mothers of daughters in 1.5-child areas were exempt from the penalty. These magnitudes are stable across estimators, and Kleibergen-Paap F-statistics range from 117.32 to 136.77 across specifications, well above conventional weak-instrument thresholds. Appendix Table E.2 confirms that instrument relevance is robust to progressively adding marital status, health insurance coverage, household asset quartiles, and quadratic cohort trends as controls (F-statistic > 100 in all specifications).

Table 4: IV Comparison: Effect of Fertility on Cognition

	(1) Between	(2) Pooled	(3) CRE	(4) First Obs.
<i>Panel A: Second stage (dep. var: cog_std)</i>				
F	-0.017 (0.147)	-0.032 (0.130)	-0.055 (0.129)	-0.049 (0.158)
<i>Panel B: First stage (dep. var: F)</i>				
girl	0.045*** (0.015)	0.039*** (0.014)	0.038*** (0.014)	0.046*** (0.015)
(1-girl) × fine	-0.105*** (0.012)	-0.121*** (0.013)	-0.121*** (0.013)	-0.107*** (0.013)
FE	County	Wave+County	Wave+County	County
Clustering	Robust	Individual	Individual	Robust
Mundlak	No	No	Yes	No
KP F	123.42	136.24	136.77	117.32
N	4116	11062	11062	4116

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

Turning to the cognition outcomes, Panel A of Table 4 reports the second-stage coefficient of fertility on the standardized measure of cognition across all four estimators. The estimated effects on cognition are -0.017 SD (s.e. 0.147) in the between specification, -0.032 SD (s.e. 0.130) in the pooled IV, -0.055 (s.e. 0.129) in the correlated random effects model, and -0.049 SD (s.e. 0.158) when using the first nonmissing observation within an individual. The lack of significance in the coefficients may be due to a lack of power, but the estimates are stable and negative across all specifications.

Tables 5 and 6 report the pooled IV estimates across outcomes and compare these to the corresponding OLS estimates. Focusing first on cognitive health (Table 5), the point estimate for the effect of a second child on standardized cognition suggests a modest reduction of 3.2 percent of a standard deviation in the preferred IV specification (Column 1). This causal estimate is even more modest than the uninstrumented OLS estimate, which suggests a reduction of 5.7 percent of a standard deviation. While the IV estimate is statistically insignificant—which might partly be due to a lack of statistical power given the sample size—the magnitude indicates that if an effect exists, it is economically modest. I find similar results for the total cognitive score (Column 2: OLS -0.308 , IV -0.173 , s.e. 0.695) and the clinical impairment cutoff (Column 3: OLS 0.023, IV 0.005, s.e. 0.059).

Table 5: Pooled: OLS and IV — Cognition Outcomes

	(1)	(2)	(3)
	Cognition (std)	Cognition (total)	Cog. impairment
<i>Panel A: OLS Estimates</i>			
F	-0.057 (0.036)	-0.308 (0.191)	0.023 (0.016)
<i>Panel B: IV Estimates (2SLS)</i>			
F	-0.032 (0.130)	-0.173 (0.695)	0.005 (0.059)
Controls	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
KP F (IV)	136.24	136.24	136.24
Wu-Hausman p	0.790	0.790	0.710
N	11062	11062	11062

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies. Sample: mothers in 1.5-child areas.

Wu-Hausman p : p -value for the test of exogeneity (H_0 : OLS is consistent). KP F: Kleibergen-Paap first-stage F -statistic.

The bottom rows of Table 5 report two diagnostic statistics. The Kleibergen-Paap F -statistic of 136 confirms instrument strength well above conventional thresholds. The Wu-Hausman p -values ($p \approx 0.79$ for standardized and total cognition; $p = 0.71$ for the clinical cutoff) fail to reject the null hypothesis of exogeneity across all three outcomes, providing no statistical evidence that $\mathcal{F}_i \times age_{it}$ is endogenous conditional on the fixed effects and included covariates. The modest attenuation of IV relative to OLS is therefore better attributed to positive selection into low fertility than to severe bias in the OLS estimator.

5.2 Effect of Fertility on Other Outcomes

Beyond direct measures of cognition, I also investigate labor supply and intergenerational transfers. Studying these additional outcomes is informative for the main research question because they represent key mediating channels through which family size could indirectly influence elderly cognition. For instance, having more children might increase financial support, allowing parents to retire earlier or invest more in their health, which could in turn protect cognitive function. Alternatively, a larger family might alter labor supply in ways that affect cognitive engagement.

Columns 1-3 of Table 6 examine labor supply at both the extensive and intensive margins. Although the cohorts being studied with two or more children are in general not more likely to work (compared to women with only one child) holding other factors constant, IV estimates suggest that women who complied with the instrument are in fact more likely to work at older ages. The point estimate on the probability of working is modestly positive (0.075, s.e. 0.053), but not statistically significant due to wide standard errors. The number of observations in Columns 2 and 3 is lower

because the sample includes individuals who have already left the labor force.²¹ Labor supply at the intensive margin, measured as weeks worked in the past year (Column 2) and hours worked in the past week (Column 3), yields coefficients of -0.558 (s.e. 2.272) and -1.435 (s.e. 3.434). While their direction is negative, which would suggest that more children reduce work at the intensive margin (e.g. through support within the family), the estimates suffer from imprecision due to low statistical power. While related research identifies a negative effect of children on old-age labor supply at the extensive margin (Rao and Zhang, 2024; Wang, 2025), these studies focus on ages past retirement age, the current identification exploits different fertility policies and instruments, which expectedly describe a different Local Average Treatment Effect as the population of compliers is different. For example, Rao and Zhang (2024) exploits differential exposure to the earlier Longer-Later-Fewer campaign, which naturally affected older cohorts (and not cohorts born after 1950 like in this study).

Columns 4-6 of Table 6 examine intergenerational transfers. A second child increases the probability of receiving any financial transfer from children or grandchildren by 61.4 percentage points. The probability that a child coresides in the same city or county rises by 21.0 percentage points, and the transfer amount increases by 3300 Yuan per year. These coefficients indicate that a second child substantially raises the material support parents receive from more children, and these estimates have causal interpretation.

Now comparing OLS and IV estimates on these outcomes within the paired table, the coefficients of labor supply are more pronounced in OLS: the point estimate on weeks worked is roughly twice as large in magnitude (about -1.28) compared to the IV result (-0.56). OLS estimates on transfers are smaller: mothers with two or more children are about 25 percentage points more likely to receive any transfer from their children (compared to IV results of 61 percentage points); the coefficient on coresidence is smaller, and the transfer amount is lower (about 1821 Yuan versus 3300 Yuan in the IV estimates). These results together suggest that compliers benefitted from having a firstborn girl, as the endogenous decision to proceed to a second child increased the level of support received, much more than correlational estimates would suggest.

²¹In China, there is no discontinuity at a statutory retirement age for rural individuals; rather, labor supply at the extensive margin dissipates smoothly at older ages, as Supplementary Appendix Figure S.1 illustrates.

Table 6: Pooled: OLS and IV — Labor Supply and Transfers

	(1)	(2)	(3)	(4)	(5)	(6)
	Working	Weeks worked	Hours/week	Lives near child	Any transfers	Transfers
<i>Panel A: OLS Estimates</i>						
F	0.014 (0.015)	-1.279* (0.689)	-0.777 (1.035)	0.119*** (0.015)	0.254*** (0.015)	1821.148*** (201.696)
<i>Panel B: IV Estimates (2SLS)</i>						
F	0.075 (0.053)	-0.558 (2.272)	-1.435 (3.434)	0.210*** (0.052)	0.614*** (0.058)	3299.376*** (839.087)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
KP F (IV)	135.50	133.92	132.68	133.44	133.34	133.25
Wu-Hausman p	0.182	0.706	0.815	0.025	0.000	0.042
N	10974	7775	7668	10960	10833	10825

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies. Sample: mothers in 1.5-child areas.

Wu-Hausman p : p -value for the test of exogeneity (H_0 : OLS is consistent). KP F: Kleibergen-Paap first-stage F -statistic.

5.3 Robustness and Additional Results

I have provided evidence for robustness of the estimated coefficients across specifications. To further show that the results are reliable, I provide additional results based on falsification tests and an alternative approach to measuring cognitive decline with individual fixed-effects.

5.3.1 Pre-OCP Falsification Test

The main results presented focus on cohorts born after 1950. This sample selection was made due to economic intuition, since the decision to proceed to a second child was not influenced by policy for cohorts born before 1950 (as I have shown above). To nevertheless show that this sample selection is appropriate, I compare first stage results using the same specification on cohorts born before 1950 and after 1950 (since cohorts born before 1950 were already at least 29 at the OCP introduction in 1979 and most would already have given birth to a second child by this age).

Table 7 compares the first stage for pre-OCP cohorts (born 1930-1949), who completed child-bearing before the introduction of fines, with post-OCP cohorts (born 1950–1979). Among pre-OCP mothers, the girl coefficient is -0.007 (s.e. 0.008) with province fixed effects and -0.008 (s.e. 0.009) with county fixed effects, which are statistically indistinguishable from zero. The fine interaction term is -0.263 (s.e. 0.188) and -0.333 (s.e. 0.204), respectively, and also statistically insignificant. The estimated joint Kleibergen-Paap F -statistics of 1.30 and 1.71 confirm that the instruments lack predictive power for these cohorts, as expected given that 96.2% of pre-OCP mothers had at least two children. By contrast, for post-OCP cohorts, both instruments are highly significant at the 1% level, with F -statistics of 135.12 and 144.87. This pattern is consistent with the instruments

operating through the policy-induced fertility constraint rather than through secular changes in son preference.

A potential concern is that the instrument’s rising predictive power reflects changing cultural preferences. However, son preference has existed historically in China. For pre-transition cohorts, it was simply dormant at the second parity because parents continued childbearing regardless of the sex of their first child. The instrument (Z_i) thus only becomes relevant as fertility constraints tighten, forcing parents to make stopping decisions at lower parities.

Table 7: Falsification Test: First Stage, Pre-OCP vs Post-OCP Cohorts

	(1)	(2)	(3)	(4)
	Pre-OCP (1930–1949)		Post-OCP (1950–1979)	
	Prov. FE	County FE	Prov. FE	County FE
girl	-0.007 (0.008)	-0.008 (0.009)	0.035*** (0.013)	0.036*** (0.013)
(1-girl) × fine	-0.263 (0.188)	-0.333 (0.204)	-0.122*** (0.012)	-0.118*** (0.012)
Controls	Yes	Yes	Yes	Yes
KP F-stat (joint)	1.30	1.71	135.12	144.87
F-stat (fine_boy only)	1.95	2.66	100.18	98.79
p-val (fine_boy)	[0.163]	[0.103]	[0.000]	[0.000]
N	5206	5206	13521	13521
Mean(F)	0.962	0.962	0.843	0.843
Mean birth year	1942.6	1942.6	1959.4	1959.4

Robust standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

5.3.2 Cognitive Decline

While the main analysis focuses on the causal effect of fertility on the level of cognition, family size might instead alter the trajectory of cognitive aging. Following [Mazzonna and Peracchi \(2024\)](#), I model cognitive decline using an individual fixed-effects specification where the parameter of interest is the interaction between fertility and age:

$$Y_{it} = \alpha_i + \tau_D(\mathcal{F}_i \times \text{Age}_{it}) + \mathbf{X}_{it}'\boldsymbol{\beta} + \theta_t + \varepsilon_{it} \quad (3)$$

Individual fixed effects (α_i) absorb all time-invariant individual heterogeneity. This removes both realized baseline fertility ($\mathcal{F}_i$) and unobserved confounding differences between individuals, such as innate cognitive endowment. Consequently, the coefficient τ_D isolates whether having additional children accelerates or decelerates the rate of cognitive decline as individuals age.

The cognitive trajectories by firstborn gender and number of children are visualized in Figures 5 and 6. While the trajectories of cognitive health do not differ much based on the gender of the firstborn, they differ substantially based on the number of children among younger cohorts (as few older cohorts have only one child). Figure 6 shows that cognition is better among individuals with only one child and the cognitive decline among these is slower. However, these individuals with one child are a selected group, highlighting the need of controlling for individual fixed effects in this section.

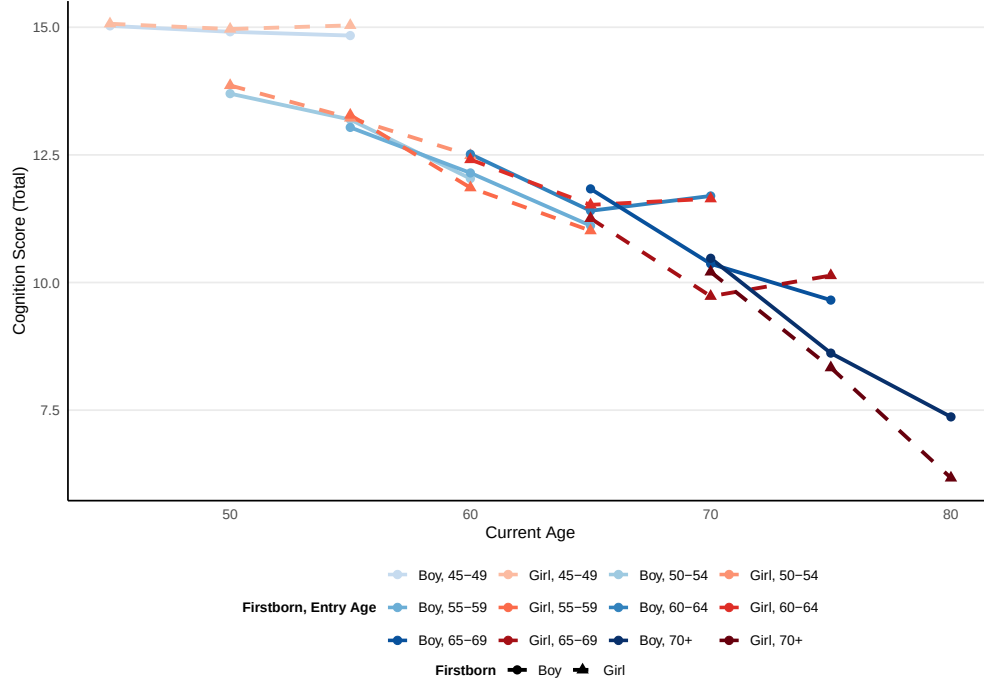


Figure 5: Cognitive Trajectories by Firstborn Gender and Entry-Age Cohort

Notes: The figure shows mean cognitive trajectories (total cognition score) by survey entry-age cohort (5-year groups). Blue lines indicate mothers with a firstborn boy, red lines mothers with a firstborn girl; darker shades represent older entry cohorts. Only groups with at least 50 observations are shown. Sample: mothers born 1930–1974 present at the 2011 baseline wave.

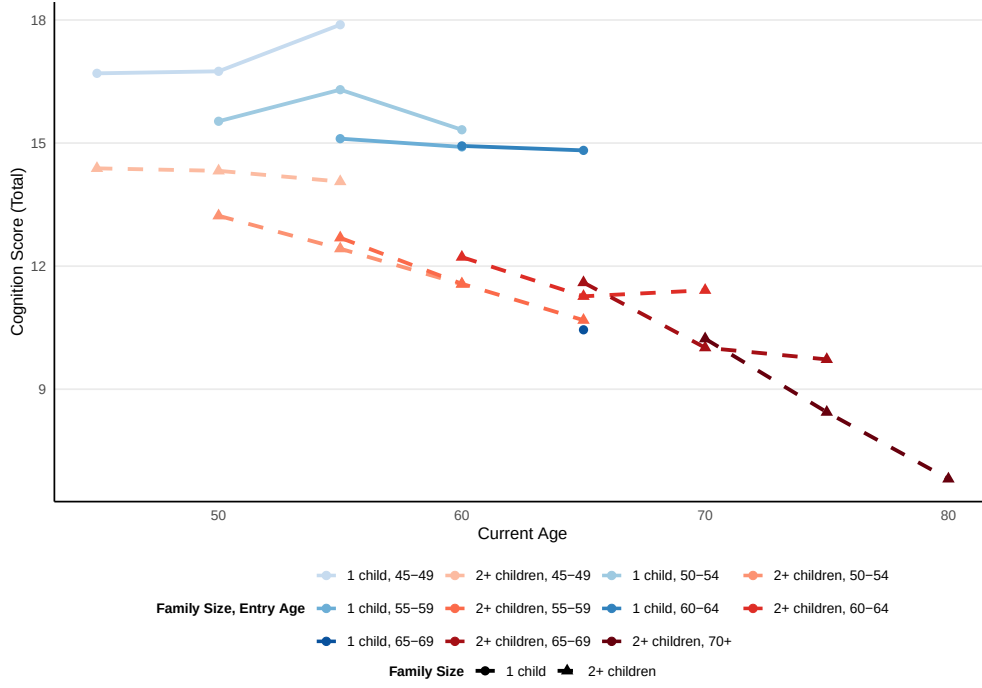


Figure 6: Cognitive Trajectories by Family Size and Entry-Age Cohort

Notes: The figure shows mean cognitive trajectories (total cognition score) by survey entry-age cohort (5-year groups). Blue lines indicate mothers with one child, red lines mothers with two or more children; darker shades represent older entry cohorts. Only groups with at least 50 observations are shown. Sample: mothers born 1930–1974 present at the 2011 baseline wave.

Since $\mathcal{F}_i$, Z_i , and the fine instrument are all time-invariant, they are absorbed by α_i and cannot serve as instruments directly. Identification of τ_D instead comes from interacting the instruments with age: the excluded instruments are $Z_i \times \text{Age}_{it}$ and $[(1 - Z_i) \cdot \frac{1}{n} \sum_{k=0}^n \text{fine}_{p,a+k}] \times \text{Age}_{it}$, which survive the within-transformation because Age_{it} varies across waves within individuals.

Table 8 reports results for the sample of mothers observed in at least two waves ($N = 10,367$). The FE-OLS estimates of τ_D on $\mathcal{F}_i \times \text{Age}_{it}$ are -0.012 (s.e. 0.006) for standardized cognition and -0.065 (s.e. 0.030) for the total score, both statistically significant at the 5% level. This indicates that fertility at the second margin accelerates cognitive decline. Specifically, the coefficient of -0.012 implies that over a 10-year span, mothers with two or more children experience an additional 0.12 SD of cognitive decline compared to mothers with one child, after controlling for time invariant individual unobserved heterogeneity.

Because $\mathcal{F}_i \times \text{Age}_{it}$ is endogenous, individual fixed effects remove only level differences of unobservables between mothers but cannot address the possibility that mothers who selected into higher fertility also differ in their rate of cognitive decline with age. The corresponding FE-IV estimates, which instrument $\mathcal{F}_i$ to isolate exogenous variation in the interaction term, are larger in magnitude at -0.015 (s.e. 0.031) and -0.082 (s.e. 0.167), respectively, but are statistically insignificant

for the continuous measures as wider confidence intervals make them imprecise. For the clinical impairment indicator, the IV coefficient is 0.029 (s.e. 0.016, $p < 0.10$).

The IV remains valid and strong, with a Kleibergen-Paap F-statistic of 51.95. Unlike in the cross-sectional setting where IV attenuates OLS estimates, the FE-IV point estimates here increase in absolute magnitude compared to FE-OLS. This amplification indicates that the FE-OLS estimates are biased toward zero. The loss of statistical significance in the continuous scores reflects the variance inflation inherent to IV estimation rather than an absence of effect. Because the IV estimates validate the direction and amplify the magnitude of the OLS results, the findings suggest that higher fertility at the second parity causally accelerates cognitive decline by more than OLS would suggest.

Table 8: Fixed Effects: OLS and IV — Effect on Cognitive Decline (Slope)

	(1) Cognition (std)	(2) Cognition (total)	(3) Cog. impairment
<i>Panel A: FE-OLS Estimates</i>			
$\mathcal{F} \times \text{Age}$	-0.012** (0.006)	-0.065** (0.030)	0.005* (0.003)
<i>Panel B: FE-IV Estimates</i>			
$\mathcal{F} \times \text{Age}$	-0.015 (0.031)	-0.082 (0.167)	0.029* (0.016)
Individual FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
Clustering	Individual	Individual	Individual
N	10367	10367	10367
KP F -stat	51.95	51.95	51.95

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sample: mothers in 1.5-child areas with ≥ 2 observations.

5.3.3 Multiple Hypothesis Testing

Because I examine a wide range of outcomes, including various measures of cognition, labor supply, and intergenerational transfers, there is a risk that some estimates appear statistically significant purely by chance. To address this, I adjust the p-values to account for the total number of hypotheses tested. Table 9 reports these adjustments using two common approaches: the Holm-Bonferroni correction and the Benjamini-Hochberg procedure. The first method is highly conservative, designed to strictly minimize the chance of making even a single false claim of significance. The second approach is slightly less restrictive, focusing instead on keeping the proportion of false

discoveries low.

The results confirm that the findings are robust to these corrections. The positive effects of having a second child on receiving financial transfers and co-residence remain statistically significant at the 1% level, even under the most conservative adjustment. This indicates that the link between family size and increased material support is not a statistical artifact. The estimates for cognitive health and labor supply remain statistically indistinguishable from zero.

Table 9: Pooled IV: Multiple Hypothesis Testing Adjustments

Outcome	Estimate (SE)	Naive p	Holm (FWER) p -value	BH (FDR) q -value
Cognition (std)	-0.032 (0.130)	0.804	1.000	0.907
Cognition (total)	-0.173 (0.695)	0.804	1.000	0.907
Cog. impairment	0.005 (0.059)	0.927	1.000	0.927
Working	0.075 (0.053)	0.162	0.970	0.364
Weeks worked	-0.558 (2.272)	0.806	1.000	0.907
Hours/week	-1.435 (3.434)	0.676	1.000	0.907
Any transfers	0.614*** (0.058)	0.000	0.000	0.000
Lives near child	0.210*** (0.052)	0.000	0.000	0.000
Transfers (level)	3299.376*** (839.087)	0.000	0.001	0.000

Notes: Standard errors clustered at individual level.

Holm (FWER) controls the Family-Wise Error Rate.

BH (FDR) controls the False Discovery Rate.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (Naive).

6 Discussion and Conclusion

This paper examines the causal effect of fertility on cognitive health in old age, exploiting quasi-exogenous variation from the sex of the firstborn child and provincial fine rates for unauthorized births under China's One-Child Policy. The identification strategy isolates compliers at the second-child margin: mothers in 1.5-child policy areas for whom the sex of their firstborn was the binding determinant of whether they proceeded to a second birth.

I estimate a Pooled IV LATE of -0.032 , suggesting a 3.2 percent reduction of a standard deviation among compliers who chose to proceed to a second child as a result of the treatment assignment. This is a local average treatment effect, and therefore does not necessarily extend to other child parities or contexts. The estimated effect is smaller in magnitude compared to the corresponding OLS estimate of -0.057 . This attenuation pattern is consistent with positive selection into low fertility: mothers who chose to have only one child are, on average, more cognitively endowed, inflating the raw uninstrumented association between higher fertility and worse cognition. Although the point estimates are stable across alternative estimators and measurements of cognition, they do not yield statistical significance. Because the confidence intervals remain wide, I fail to reject the null hypothesis for the baseline level of cognition, though I cannot definitively rule out larger causal effects.

I also provide complementary evidence with a fixed-effects IV specification that controls for individual-specific unobserved heterogeneity. Since the instruments do not vary within individuals, I apply this method to study the rate of cognitive decline, rather than the baseline level of cognitive health. Individual fixed effects remove only time-invariant level differences between mothers and cannot account for the possibility that mothers who selected into higher fertility also differ in their unobserved rate of cognitive aging. The fixed-effects OLS estimates of the $\mathcal{F}_i \times \text{Age}_{it}$ interaction are statistically significant, suggesting a faster cognitive decline among mothers with two or more children. The instrumented FE-IV point estimates increase in absolute magnitude, confirming that the FE-OLS estimates are biased toward zero and that the causal impact of higher fertility at the second margin on the rate of cognitive decline is larger than OLS suggests. Together, these results suggest that while I do not detect a significant worsening of the baseline *level* of cognition, higher fertility at the second margin causally accelerates the *pace* of cognitive decline as mothers age.

These findings stand in contrast to [Bonsang and Skirbekk \(2022\)](#), who identify a negative causal effect of a *third* child on the level of cognition among European adults. Several differences in context and design may explain the divergence. First, the relevant margin differs: I study the second-child margin in a setting shaped by strict fertility policy, whereas [Bonsang and Skirbekk \(2022\)](#) identify effects at the third parity in contexts without comparable restrictions. Second, the Chinese elderly depend heavily on children for old-age support in the absence of a comprehensive pension system. In this environment, an additional child may generate offsetting cognitive benefits through social engagement, caregiving, and material transfers that are less available in European welfare-state contexts. Third, the instruments differ in the population they identify: sex-composition instruments identify mothers with a preference for mixed-sex sibships, who may differ systematically from

compliers with strong son preference driven by policy enforcement in rural China. The null result in Sweden by [Fischer et al. \(2025\)](#), who find no causal impact on dementia using administrative data, and the positive effect found by [Bai et al. \(2025\)](#) at low parities in China using a broader policy exposure variation, together point to a parity- and context-specific relationship between fertility and late-life cognition that cannot be easily generalized.

The transfer results reinforce this interpretation. A second child causally increases the probability of receiving financial support from children by 61.4 percentage points, the likelihood of co-residence by 21.0 percentage points, and the annual transfer amount by approximately 3,300 Yuan. These effects are considerably larger than the corresponding OLS estimates—with Wu-Hausman tests formally rejecting the consistency of OLS—and survive multiple-hypothesis corrections at the 1% level, suggesting that compliers, mothers in rural 1.5-child areas for whom son preference was a binding motive, benefit disproportionately from intergenerational support. This pattern replicates and extends the evidence in [Oliveira \(2016\)](#), who uses twin births to document positive effects of additional children on financial transfers and co-residence in China, and in [Rao and Zhang \(2024\)](#), who similarly finds that higher fertility increases material support flows using the LLF campaign as exogenous variation. Together these results suggest that at the second-child margin in rural China, additional children function primarily as old-age insurance, rather than as a source of cognitive strain.

Overall, these results suggest that while earlier estimates linking higher fertility to worse baseline *levels* of late-life cognition largely reflect positive selection into low fertility—exaggerating the uninstrumented causal relationship—having a second child does causally accelerate the *trajectory* of cognitive decline over time. This is consistent with the theoretical framework of [Guo et al. \(2025\)](#), who distinguish between wanted and unwanted fertility and show that the direction of selection bias from endogenous fertility choices depends critically on the policy environment and individual preferences. In China during the One-Child Policy period, coercive downward pressure on fertility was strongest for mothers with a preference for more children — particularly in rural 1.5-child areas with strong son preference — meaning that compliers in this study are, by construction, mothers with a preference for larger families. This context implies that selection into low fertility here is positive (cognitively endowed mothers are *not* the ones selecting into fewer children), which is indeed what the attenuation of OLS estimates under instrumentation suggests.

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APPENDIX

Appendix

Appendix A: Sample Characteristics and Sample Selection

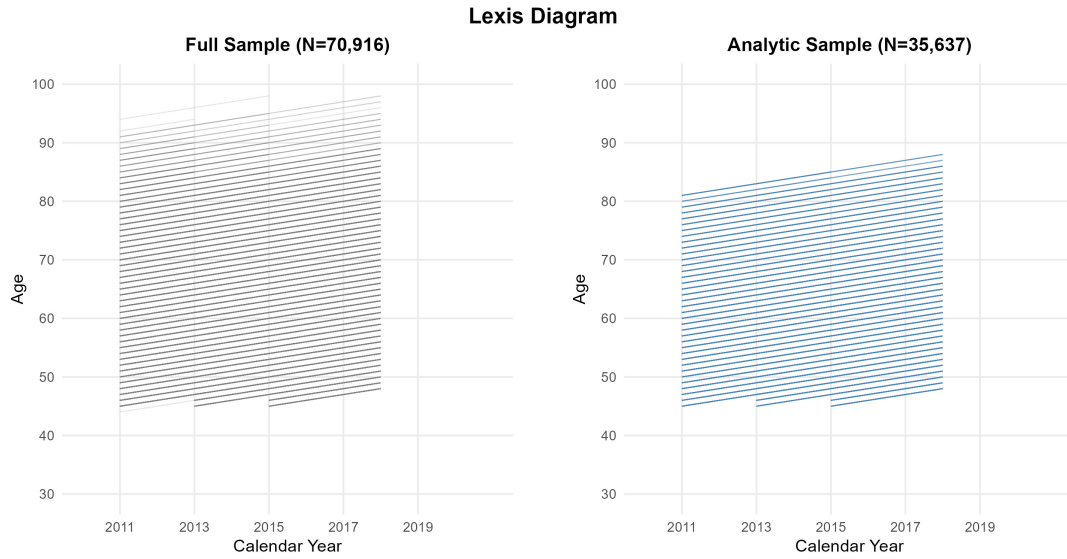


Figure A.1: Lexis diagram: Sample attrition across cohorts and ages, showing the analytic sample used in the main analysis.

Table A.1: Sample Selection Attrition: Main Sample

Step	Selection Criterion	N Remaining	N Dropped	% of Original
0	Aged 45 and older	70,916	0	100%
1	Drop birth cohort after 1974	70,916	0	100%
2	Drop birth cohort before 1930	69,916	1,000	98.6%
3	Has at least one child	69,418	498	97.9%
4	Female respondents only (mothers)	35,685	33,733	50.3%
5	Age at birth 15–45 years	35,685	0	50.3%
6	First birth before 2002	35,637	48	50.3%

Table A.2: Sample Selection Attrition: IV Analysis Restrictions

Step	Selection Criterion	N Remaining	N Dropped	% of Main
1	1.5-child policy areas only	18,996	16,641	53.3%
2	Non-missing fine instrument	18,952	44	53.2%
3	One observation per mother	5,989	12,963	16.8%

Notes: Percentages in Table B are relative to the main sample (N=35,637) from Table A.

Detailed information on the sample selection process is provided in Tables [A.1](#) and [A.2](#), which document each step of the selection criteria and the resulting sample sizes.

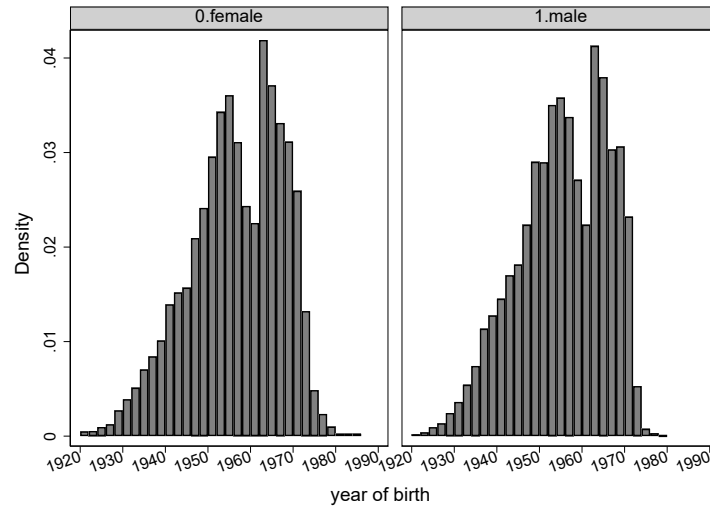


Figure A.2: Distribution of cohorts in the sample

Notes: The large drop in probability mass for individuals born during 1950-1960 is attributed to the Great Famine. CHARLS is nationally representative of the Chinese elderly aged 45 and older.



Figure A.3: Sample Selection

Notes: The figure shows the selected sample vs the full original sample.

Appendix B: Birth Information

Table B.1: Probability of second child in Urban areas by policy area

Policy Area / Firstborn Sex	Birth Cohort				
	1930	1940	1950	1960	1970
1-child					
Female	0.97	0.91	0.64	0.42	0.45
Male	0.92	0.88	0.56	0.32	0.29
<i>Difference (F-M)</i>	0.05	0.02	0.09	0.10	0.16
1.5-child					
Female	0.96	0.96	0.87	0.74	0.60
Male	1.00	0.98	0.80	0.62	0.43
<i>Difference (F-M)</i>	-0.04	-0.03	0.07	0.12	0.17
2+-child					
Female	0.67	1.00	0.57	0.77	0.71
Male	1.00	0.86	0.85	0.62	0.60
<i>Difference (F-M)</i>	-0.33	0.14	-0.27	0.14	0.11

Notes: Table shows the probability of having a second child by firstborn sex for urban mothers across all OCP policy areas. For each policy area, the Difference row shows son preference (Female - Male). Positive values indicate higher continuation probability when the firstborn is female.

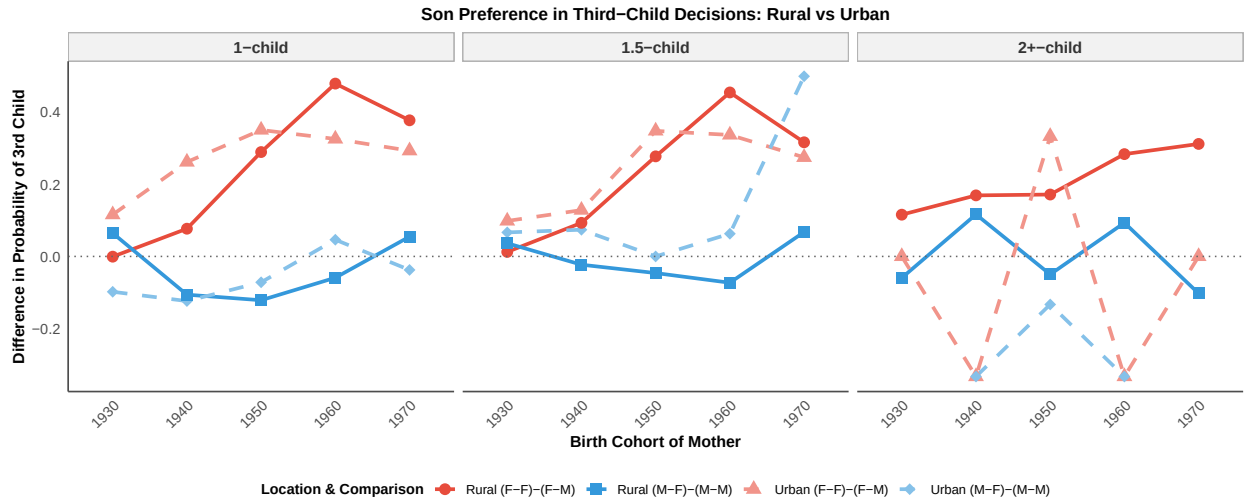


Figure B.1: Son preference in third-child decisions by policy area, rural and urban communities

Notes: The figure shows the difference in probability of having a third child across birth cohorts and OCP policy areas. Each panel represents one policy area (1-child, 1.5-child, 2+-child). Red lines show rural communities, pink lines show urban communities. Solid lines show (F-F)-(F-M) differences (son preference when firstborn is female), dashed lines show (M-F)-(M-M) differences (son preference when firstborn is male). Positive values indicate higher probability of having a third child when the previous births did not produce the desired sex composition. Rural areas exhibit substantially stronger son preference effects, particularly in 1-child policy areas where rural 1960 cohort mothers show a 48 percentage point difference compared to 33 percentage points in urban areas.

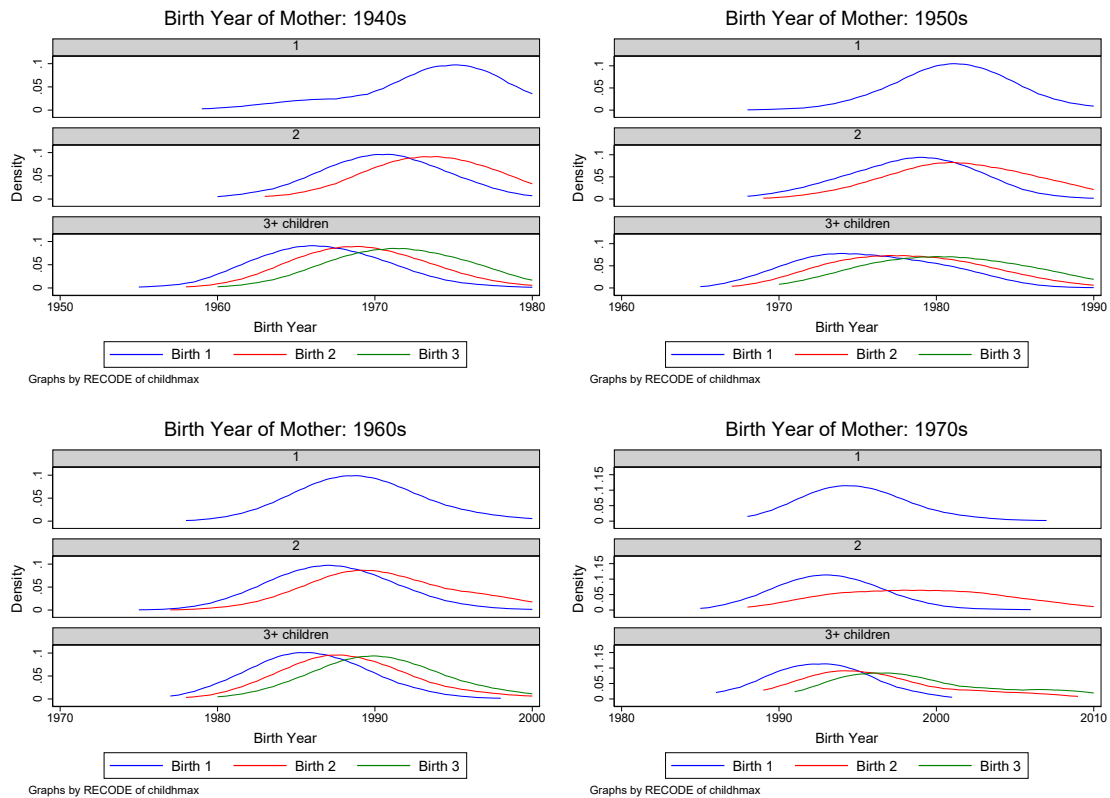


Figure B.2: Birth year and spacing of children by mothers' birth cohort

Table B.2: Probability of third child by first and second child sex (Rural vs Urban)

Policy Area	Rural					Urban				
	1930	1940	1950	1960	1970	1930	1940	1950	1960	1970
1-child										
F-F	0.97	0.94	0.81	0.80	0.60	0.89	0.83	0.64	0.45	0.36
F-M	0.97	0.87	0.52	0.32	0.22	0.77	0.57	0.29	0.12	0.07
M-F	0.96	0.86	0.53	0.31	0.32	0.73	0.54	0.34	0.22	0.06
M-M	0.89	0.97	0.65	0.37	0.26	0.83	0.66	0.41	0.17	0.10
<i>Diff: (F-F)-(F-M)</i>	-0.00	0.08	0.29	0.48	0.38	0.12	0.26	0.35	0.33	0.29
<i>Diff: (M-F)-(M-M)</i>	0.06	-0.11	-0.12	-0.06	0.05	-0.10	-0.12	-0.07	0.05	-0.04
1.5-child										
F-F	0.96	0.97	0.82	0.71	0.45	0.92	0.90	0.67	0.50	0.40
F-M	0.95	0.88	0.54	0.26	0.13	0.82	0.77	0.32	0.16	0.12
M-F	0.96	0.89	0.57	0.30	0.18	1.00	0.74	0.38	0.36	0.67
M-M	0.93	0.91	0.61	0.38	0.11	0.93	0.67	0.38	0.29	0.17
<i>Diff: (F-F)-(F-M)</i>	0.01	0.09	0.28	0.46	0.32	0.10	0.13	0.35	0.34	0.28
<i>Diff: (M-F)-(M-M)</i>	0.04	-0.02	-0.05	-0.07	0.07	0.07	0.07	0.00	0.06	0.50
2+-child										
F-F	0.94	0.98	0.71	0.61	0.42	1.00	0.67	0.50	0.00	0.00
F-M	0.82	0.81	0.53	0.32	0.11	1.00	1.00	0.17	0.33	0.00
M-F	0.89	0.92	0.58	0.40	0.19	–	0.67	0.20	0.00	–
M-M	0.95	0.80	0.63	0.30	0.30	1.00	1.00	0.33	0.33	0.00
<i>Diff: (F-F)-(F-M)</i>	0.12	0.17	0.17	0.28	0.31	0.00	-0.33	0.33	-0.33	0.00
<i>Diff: (M-F)-(M-M)</i>	-0.06	0.12	-0.05	0.09	-0.10	–	-0.33	-0.13	-0.33	–

Notes: Table shows the probability of having a third child by the sex of the first two children for mothers across all OCP policy areas, split by rural and urban communities. F = Female first-born/secondborn, M = Male firstborn/secondborn. Each policy area shows four sex combinations (F-F, F-M, M-F, M-M) plus two difference rows. The first difference row (F-F)-(F-M) shows son preference when the firstborn is female (positive values indicate higher third-child probability when both first two children are female). The second difference row (M-F)-(M-M) shows son preference when the firstborn is male (positive values indicate higher third-child probability when the firstborn is male and secondborn is female). Rural areas generally show stronger son preference effects than urban areas, particularly in 1-child policy areas among 1960-1970 cohorts.

	rural/urban (village head)									
	0.community cohort group (10-year)					1.village cohort group (10-year)				
	1930	1940	1950	1960	1970	1930	1940	1950	1960	1970
Mean										
age at 1st birth	22.7	23.6	24.4	24.0	23.7	22.4	22.3	23.0	22.6	22.6
age at 2nd birth	26.1	26.1	26.6	27.0	28.3	26.3	24.8	25.7	25.8	27.5
age at 3rd birth	28.5	27.9	28.0	27.2	28.1	29.2	27.2	27.5	27.0	29.0
ageatbirthlast	33.3	29.9	28.0	26.5	26.6	37.5	31.8	29.0	26.9	27.5
Standard deviation										
age at 1st birth	4.1	3.7	3.4	3.3	3.6	4.1	3.7	3.4	2.7	2.8
age at 2nd birth	4.7	4.0	4.2	5.0	5.7	5.7	3.9	4.0	4.1	5.1
age at 3rd birth	4.8	3.5	4.9	4.7	5.8	5.7	4.3	4.2	4.0	5.6
ageatbirthlast	7.0	4.5	4.2	4.3	4.8	8.2	5.6	4.8	4.4	5.4
Median										
age at 1st birth	22.0	23.0	24.0	24.0	24.0	22.0	22.0	23.0	22.0	22.0
age at 2nd birth	25.0	26.0	26.0	26.0	28.0	25.0	24.0	25.0	25.0	27.0
age at 3rd birth	28.0	28.0	27.0	26.0	28.0	29.0	27.0	27.0	26.0	28.0
ageatbirthlast	33.0	29.0	27.0	25.0	25.0	37.0	31.0	28.0	26.0	26.5
Number of nonmissing values										
age at 1st birth	1172.0	2536.0	4516.0	4416.0	928.0	2584.0	6248.0	10592.0	11198.0	2132.0
age at 2nd birth	1116.0	2312.0	2972.0	1992.0	375.0	2536.0	6092.0	9764.0	9261.0	1561.0
age at 3rd birth	924.0	1596.0	1308.0	534.0	57.0	2384.0	5540.0	6232.0	3962.0	386.0
ageatbirthlast	1172.0	2536.0	4516.0	4416.0	928.0	2584.0	6248.0	10592.0	11198.0	2132.0

Table B.3: Age at birth at each parity, by rural/urban

	Policy Han (2011)									
	1 cohort group (10-year)					(1.5/2+) cohort group (10-year)				
	1930	1940	1950	1960	1970	1930	1940	1950	1960	1970
Mean										
age at 1st birth	22.8	23.1	24.0	23.7	23.5	22.3	22.3	23.1	22.6	22.6
age at 2nd birth	26.5	25.6	26.4	26.7	28.1	26.1	24.8	25.7	25.8	27.5
age at 3rd birth	29.2	27.6	27.9	27.8	30.0	28.9	27.2	27.4	26.8	28.5
ageatbirthlast	34.9	30.4	28.3	26.7	26.9	37.2	31.8	29.0	26.9	27.3
Standard deviation										
age at 1st birth	4.0	3.6	3.4	3.3	3.3	4.2	3.8	3.4	2.7	2.9
age at 2nd birth	5.4	3.7	4.2	4.6	4.9	5.5	4.1	3.9	4.1	5.3
age at 3rd birth	6.2	3.7	4.5	4.2	5.0	5.0	4.4	4.3	4.0	5.7
ageatbirthlast	8.3	4.5	4.3	4.4	4.9	7.8	5.8	4.8	4.4	5.4
Median										
age at 1st birth	22.0	23.0	24.0	23.0	23.0	22.0	22.0	23.0	23.0	22.0
age at 2nd birth	26.0	25.0	26.0	26.0	28.0	25.0	24.0	25.0	25.0	27.0
age at 3rd birth	28.0	27.0	27.0	27.0	30.0	28.0	27.0	27.0	26.0	27.0
ageatbirthlast	34.0	30.0	27.5	26.0	26.0	37.0	31.0	28.0	26.0	26.0
Number of nonmissing values										
age at 1st birth	1508.0	3456.0	5600.0	5301.0	1063.0	2220.0	5236.0	9272.0	10094.0	1946.0
age at 2nd birth	1444.0	3220.0	4052.0	2822.0	499.0	2184.0	5104.0	8488.0	8269.0	1399.0
age at 3rd birth	1240.0	2492.0	2168.0	1005.0	116.0	2048.0	4572.0	5260.0	3426.0	316.0
ageatbirthlast	1508.0	3456.0	5600.0	5301.0	1063.0	2220.0	5236.0	9272.0	10094.0	1946.0

Table B.4: Age at birth at each parity, by policy area

Children by Sex of Firstborn - Cohort 1970-1979

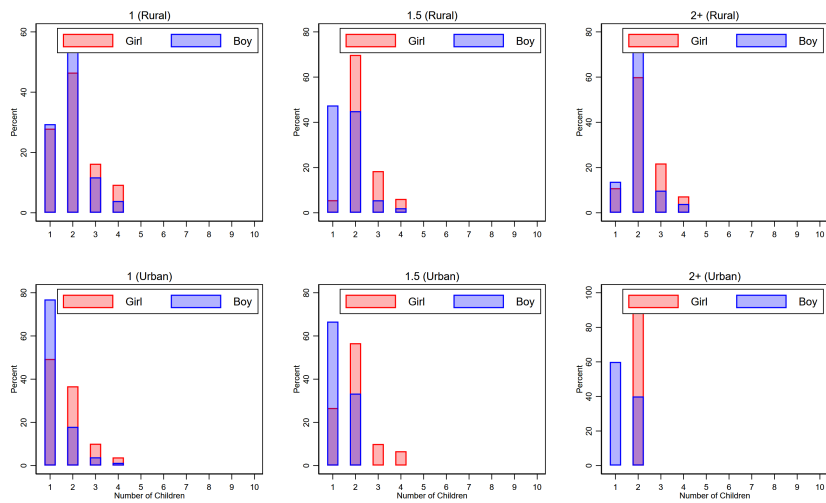


Figure B.3: Distribution of children by cohort (1970 birth cohort)

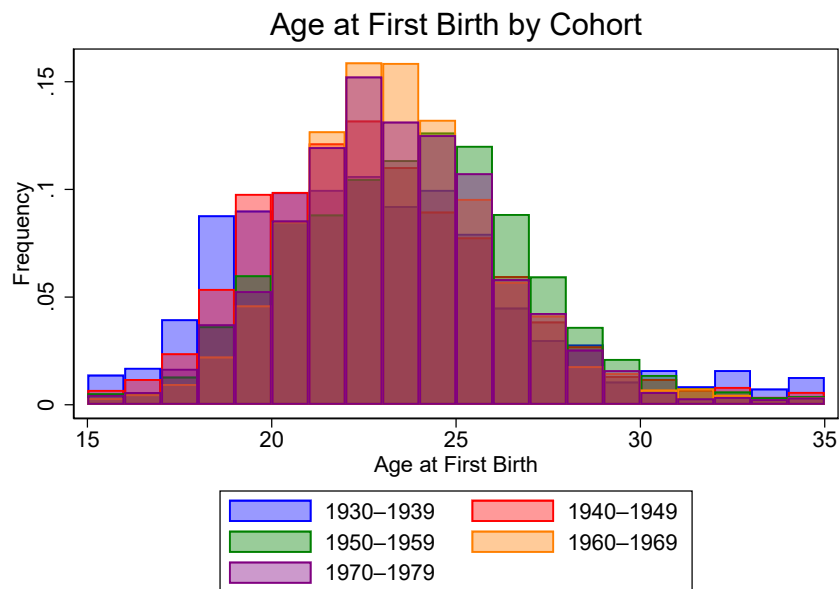


Figure B.4: Age at first birth by birth cohort of mother.

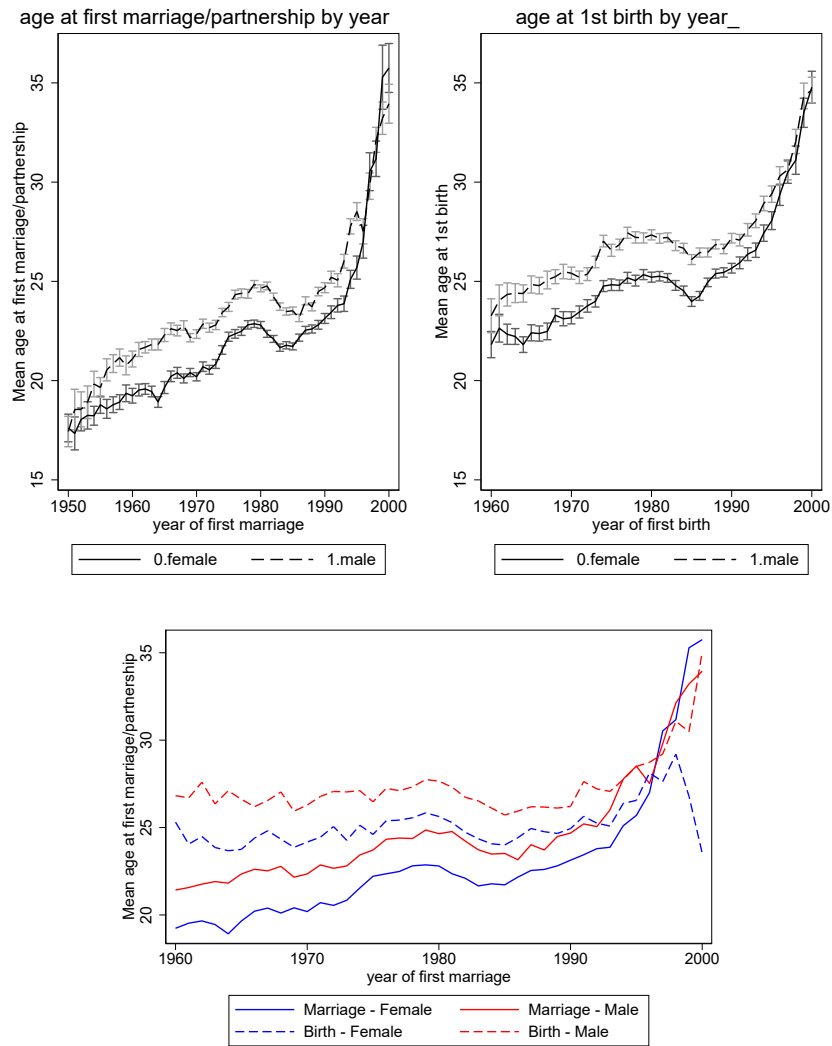


Figure B.5: Age at first marriage and first birth by year

Notes:

Appendix C: Construction of cognition score

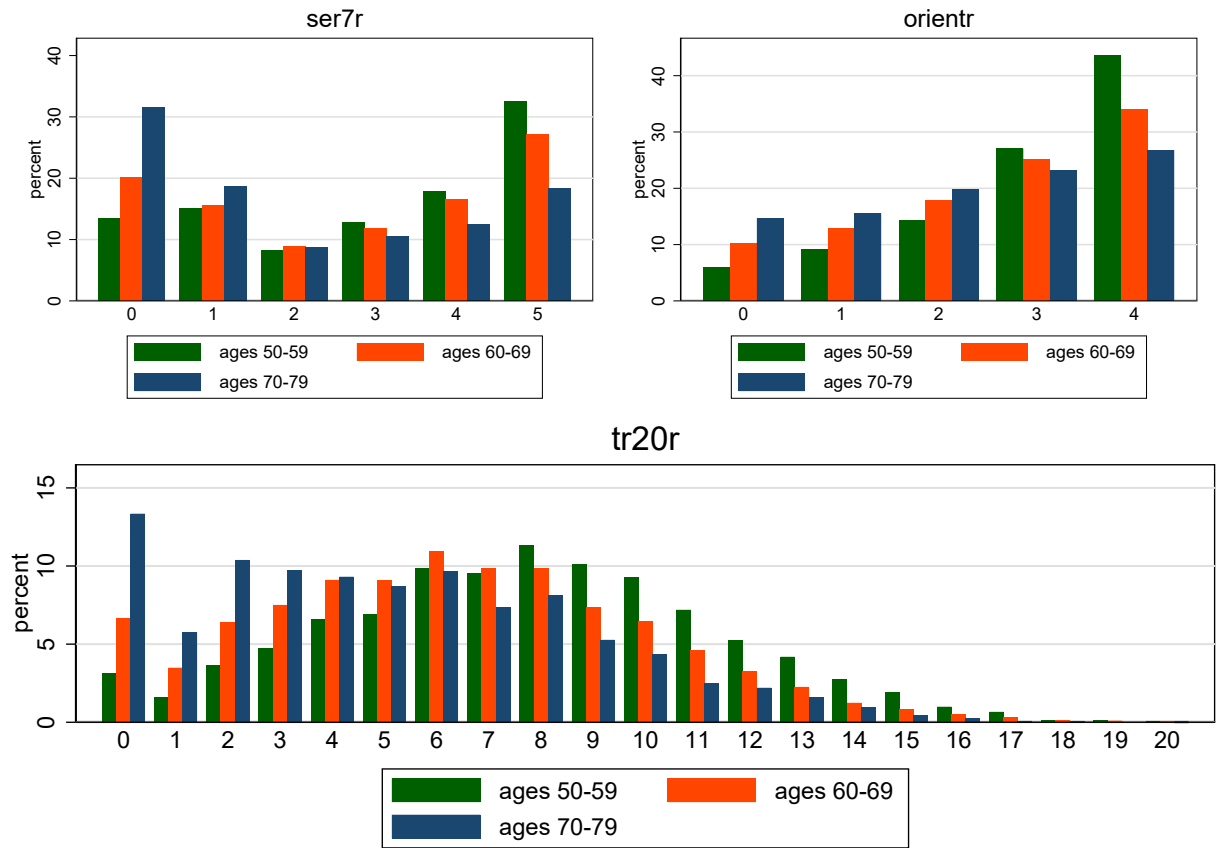


Figure C.1: Cognition Scales by age-group

	current-age-group (10-year)				
	ages 45-49	ages 50-59	ages 60-69	ages 70-79	ages 80+
Mean					
r Self-reported memory	3.94	4.10	4.15	4.19	4.20
r recall summary score	8.74	7.94	6.61	5.39	3.69
r serial 7s	3.67	3.41	3.23	2.74	1.87
r cognition orient (summary date)	3.24	3.07	2.84	2.60	2.12
r immediate word recall	4.52	4.11	3.52	2.95	2.05
r delayed word recall	4.19	3.80	3.05	2.39	1.55
r able to draw picture	0.77	0.70	0.59	0.50	0.30
Cognition Total	15.80	14.64	12.99	11.17	8.33
Cognition (Z-score)	0.40	0.19	-0.12	-0.46	-0.99
Number of nonmissing values					
r Self-reported memory	10,483	22,940	20,405	9,051	2,059
r recall summary score	9,948	21,555	18,675	7,852	1,533
r serial 7s	10,139	21,935	19,314	8,344	1,771
r cognition orient (summary date)	10,362	22,522	19,908	8,722	1,934
r immediate word recall	10,075	21,878	19,092	8,102	1,611
r delayed word recall	9,957	21,571	18,685	7,862	1,551
r able to draw picture	10,250	22,270	19,206	8,336	1,730
Cognition Total	9,622	20,549	17,670	7,283	1,351
Cognition (Z-score)	9,622	20,549	17,670	7,283	1,351
in curr. wave	11,489	24,753	21,685	10,008	2,981

Table C.1: Cognition score by age

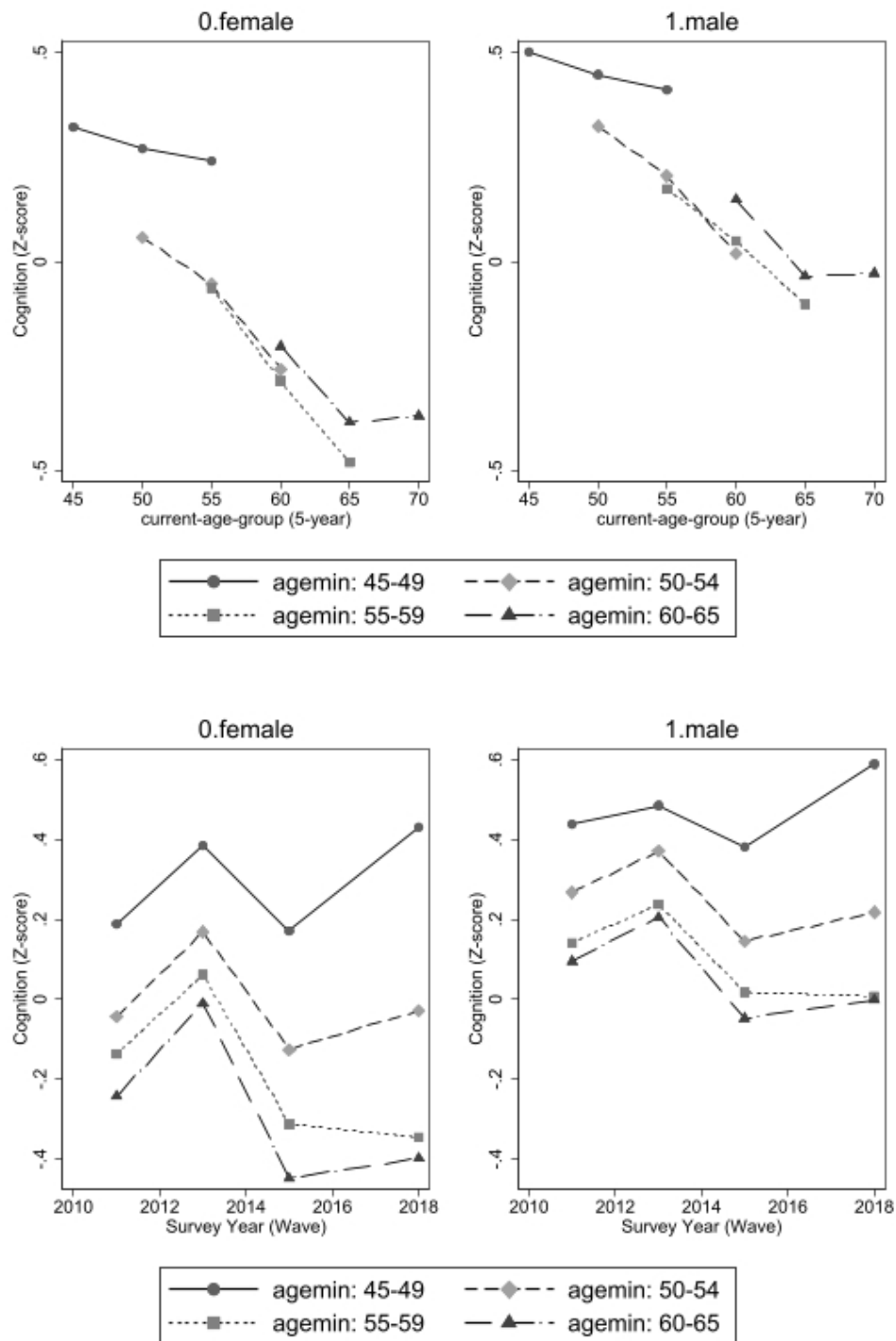


Figure C.2: Evolution of cognition over time, by age and gender, 2011–2018.

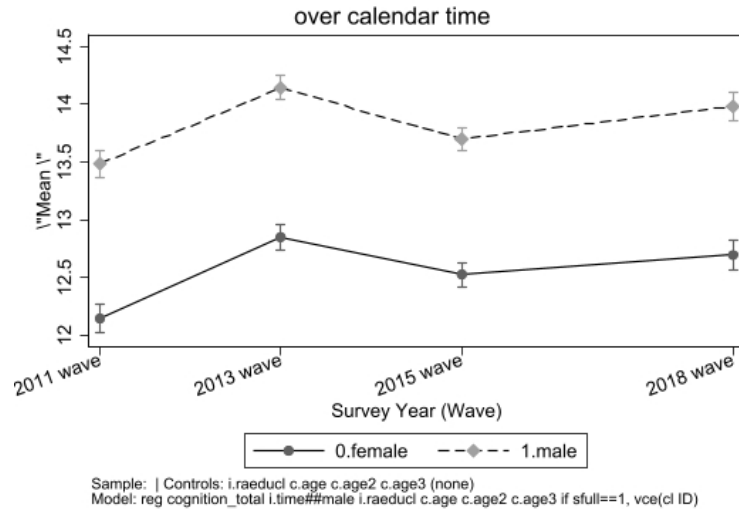


Figure C.3: Evolution of cognition over time (age-adjusted trends), 2011–2018.

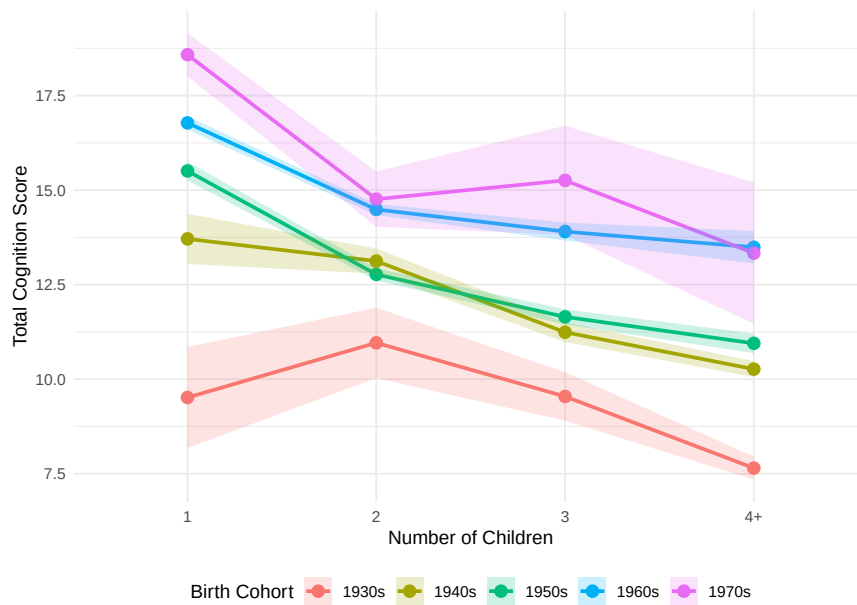


Figure C.4: Cognition Score by Fertility and Birth Cohort (All Ages)

Notes: Mean cognition scores across fertility levels (1-4+ children) with separate lines for birth cohorts (1930s-early 1970s). Shaded areas represent 95% confidence intervals. Cohorts are observed at different ages due to the 7-year window (2011-2018); future research can reevaluate as cohorts age. Sample: all mothers born 1930-1974.

Appendix D: Policy heterogeneity within and between regions and across time

Data on OCP Fines by Ebenstein (2010)

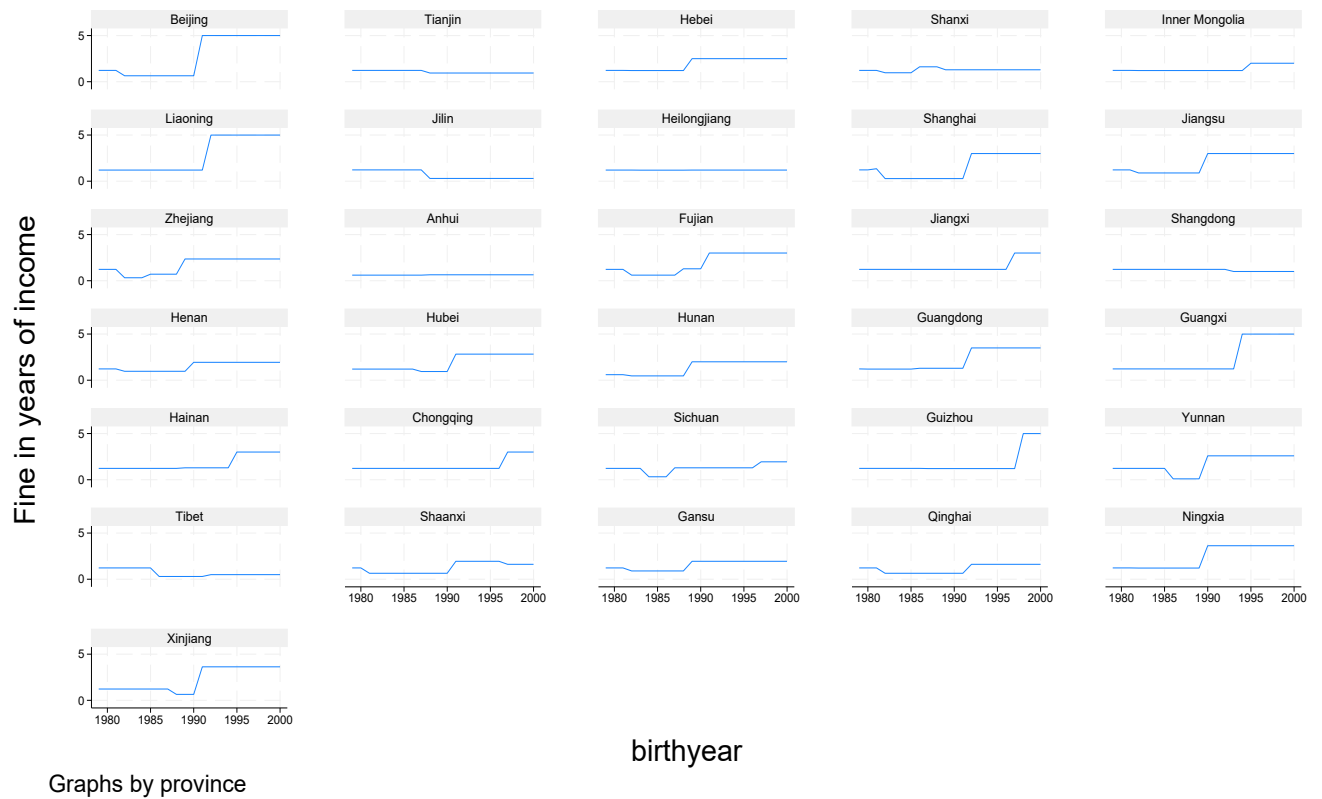


Figure D.1: Evolution of fines for above-quota births

Notes: Source: Ebenstein (2010). The figure illustrates the evolution of fines for above-quota births. Fines for children above the permitted quota significantly increased in the early 1990s in a large number of provinces.

Policy information from Community Questionnaire (reported by community-head)

Table D.1: OCP policy (community-level)

	rural or urban (village head response)		
	0.community	1.village	Total
Factor-variable proportion			
Policy for Ethnic Han (2011)=1 child	.8082	.2088	.4063
Policy for Ethnic Han (2011)=1.5 child	.1644	.6397	.4831
Policy for Ethnic Han (2011)=2 child	.0274	.138	.1016
Policy for Ethnic Han (2011)=3+	0	.01347	.009029
Frequency	150	300	450

Table D.2: OCP policy (ID-level)

	rural or urban (village head response)		
	0.community	1.village	Total
Factor-variable proportion			
Policy for Ethnic Han (2011)=1 child	.7734	.1984	.352
Policy for Ethnic Han (2011)=1.5 child	.2022	.6533	.5328
Policy for Ethnic Han (2011)=2 child	.02444	.1356	.1059
Policy for Ethnic Han (2011)=3+	0	.01267	.009283
Frequency	6,187	16,636	22,823

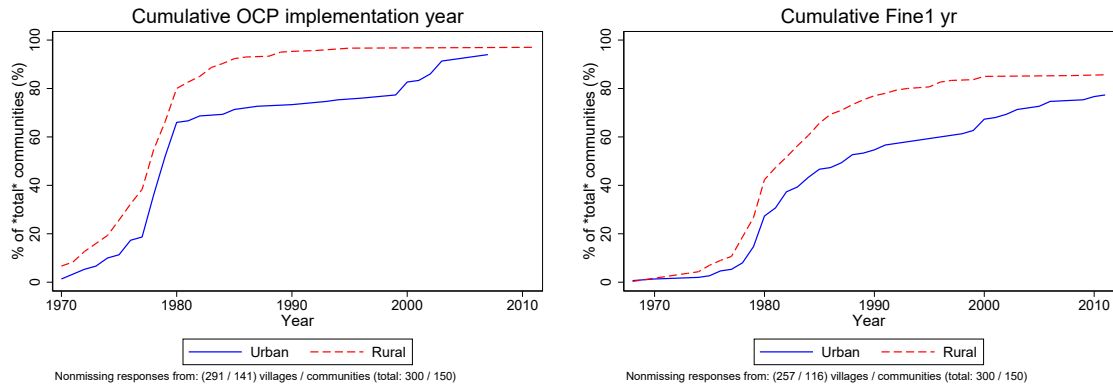


Figure D.2: First reported year of implementation of the One-Child Policy / fines

Notes: Reported by village/community heads.

	Mean										Number of nonmissing values							
	Fine1	fine2	fine3	fine4	OCF start	Fine1 yr	fine2 yr	fine3 yr	fine4 yr	Fine1	fine2	fine3	fine4	OCF start	Fine1 yr	fine2 yr	fine3 yr	fine4 yr
province																		
Beijing	15400	.	.	.	1986	1986	.	.	.	2	0	0	0	3	3	0	0	0
Tianjin	1500	1000	2000	20000	1978	1980	1980	1990	2000	2	1	1	1	3	2	1	1	1
Hebei	2291	12131	18470	39156	1978	1984	1991	1998	2004	16	10	10	10	18	18	10	10	10
Shanxi	1124	1485	3767	4283	1979	1981	1989	1998	2000	12	10	9	6	12	12	10	9	6
Inner Mongolia	5364	1080	1760	10500	1981	1984	1987	1993	2007	11	5	5	4	17	12	5	5	4
Liaoning	2762	17900	21667	31667	1981	1987	1988	1994	1993	12	4	3	3	15	13	4	3	3
Jilin	928	1470	6250	6667	1983	1978	1986	1994	2001	5	4	4	3	9	5	4	4	3
Heilongjiang	1843	5500	6500	4000	1986	1982	1994	1997	2004	7	2	2	2	12	8	2	2	2
Shanghai	.	.	.	.	1994	2006	.	.	.	0	0	0	0	3	1	0	0	0
Jiangsu	3843	1544	7622	16625	1977	1982	1988	2000	2008	15	9	9	8	20	15	9	9	8
Zhejiang	8822	12130	17091	34194	1985	1989	1992	1998	2005	15	10	9	9	19	18	11	9	9
Anhui	936	3453	6406	15400	1981	1984	1992	1998	2005	21	19	18	18	21	21	19	18	18
Fujian	3003	1045	6656	24360	1977	1982	1989	1997	2003	11	10	10	10	12	12	11	11	11
Jiangxi	912	3896	5478	12724	1980	1987	1991	1994	2002	16	12	9	9	25	20	13	10	9
Shandong	4999	4032	13947	30529	1978	1984	1989	1998	2006	27	17	15	14	32	29	17	15	14
Henan	1101	2977	4060	7642	1978	1984	1989	1996	2004	26	21	20	16	33	25	21	20	16
Hubei	2242	4726	11067	21833	1977	1985	1990	1998	2004	11	7	6	6	12	11	7	6	6
Hunan	827	3857	8560	13587	1978	1983	1993	2000	2007	17	15	15	15	18	18	16	16	16
Guangdong	3949	22538	45480	97465	1981	1984	1988	1997	2005	30	19	17	13	31	33	19	17	13
Guangxi	10436	804	15511	36389	1980	1983	1980	1995	2002	15	9	9	9	15	15	9	9	9
Chongqing	1481	1817	6000	18667	1977	1976	1984	1993	2005	6	6	6	6	6	6	6	6	6
Sichuan	532	1836	3976	13308	1977	1978	1986	1993	2003	25	18	18	16	34	27	18	18	16
Guizhou	1183	533	1233	8000	1983	1986	1990	1995	2001	6	3	3	3	6	6	3	3	3
Yunnan	5334	1595	5815	8878	1979	1987	1991	1997	2003	20	13	10	9	25	21	13	11	9
Shaanxi	381	1014	2580	4050	1979	1981	1989	1998	2003	7	7	5	4	14	7	7	5	4
Gansu	600	44000	5001500	.	1978	1982	1997	1980	1990	9	5	1	0	11	9	5	1	1
Qinghai	183	1650	3000	.	1982	1988	1999	2005	.	3	2	1	0	3	3	2	1	0
Xinjiang	400	933	6000	40000	1993	1993	1997	2001	2009	3	3	3	3	3	3	3	3	3
Total	3069	6000	33893	23971	1980	1984	1989	1997	2004	350	241	218	197	432	373	245	222	200

Table D.3: Reported Policy Fines (community head response), at each implementation

Notes: Values reported in nominal Yuan.

	rural/urban (village head)							
	0.community				1.village			
	Policy Han (1st)				Policy Han (1st)			
	1	1.5	2+	Total	1	1.5	2+	Total
province								
Beijing	3							
Tianjin	1				2			
Hebei	4		2		1	3	8	
Shanxi	2				3	5	2	
Inner Mongolia	6	1	3		2	4	2	
Liaoning	6		1		3	2	3	
Jilin	4		1		2	2		
Heilongjiang	9		1		1	1		
Shanghai	2				1			
Jiangsu	5		1		14		1	
Zhejiang	6	2			2	6	4	
Anhui	2	2			1	11	5	
Fujian						9	3	
Jiangxi	10	1	2		4	5	5	
Shandong	3	4	4		2	12	8	
Henan	7	1	1		7	8	9	
Hubei	2	1	1		1	1	6	
Hunan	2	5			4	8	2	
Guangdong	3	2	6			5	17	
Guangxi	4	1				7	2	
Chongqing	1	1				2	2	
Sichuan	5		1		19	2	9	
Guizhou	1				2	2	1	
Yunnan	4				1	2	19	
Shaanxi	5	1			1	4	4	
Gansu	3					6	3	
Qinghai							3	
Xinjiang			1					
Total				147				298

Table D.4: Policy area across sampled counties (community head response), at first implementation

Notes:

province	rural/urban (village head)							
	0.community				1.village			
	Policy Han (2011)				Policy Han (2011)			
	1	1.5	2+	Total	1	1.5	2+	Total
Beijing	3							
Tianjin	1				2			
Hebei	5		1		6		6	
Shanxi	2				2	7	1	
Inner Mongolia	8	2			8			
Liaoning	7				1	7		
Jilin	4	1			4			
Heilongjiang	10				1		1	
Shanghai	2				1			
Jiangsu	5	1			8	3	3	
Zhejiang	8	1			1	11		
Anhui	2	2			2	15		
Fujian						12		
Jiangxi	10	3			1	13		
Shandong	4	7			1	21		
Henan	7	1			7	15	1	
Hubei	3	1			1	5	2	
Hunan	4	3			2	12		
Guangdong	6	2	2		2	18	2	
Guangxi	5					10		
Chongqing	2				1	2	1	
Sichuan	5		1		22	2	6	
Guizhou	1				5			
Yunnan	4				3	2	17	
Shaanxi	6				1	7	1	
Gansu	3				1	7	1	
Qinghai							3	
Xinjiang	1							
Total				146				297

Table D.5: Policy area across sampled counties (community head response), in 2011

Notes: Compared to the policy type at the first implementation (Table D.4), community heads reported a current policy that was in most cases stricter in nearly all provinces in the sampled counties (Hebei, Jilin and Gansu in rural areas are slightly relaxed in comparison - possibly due to sampling chance)

	rural/urban (village head)							
	0.community				1.village			
	2nd policy rule				2nd policy rule			
	1	1.5	2+	Total	1	1.5	2+	Total
Policy Han (1st)								
1								
province								
Hebei							1	
Shanxi						2		
Inner Mongolia		1	1			2		
Liaoning						3		
Jilin						1		
Heilongjiang							1	
Jiangsu			1		1	4	1	
Zhejiang	1		1			2		
Anhui						1		
Jiangxi						3		
Shandong						1		
Henan						3		
Hubei						1		
Hunan						3		
Sichuan						2	1	
Guizhou							1	
Yunnan							1	
Shaanxi						1		
1.5								
province								
Shanxi					2			
Inner Mongolia		1						
Liaoning							1	
Heilongjiang					1			
Anhui		1						
Jiangxi	1							
Shandong	1							
Henan					2			
Hubei		1						
Hunan	2				1	1		
Guangdong	1				1	1		
Guangxi	1				1	2		
Chongqing	1							
Sichuan					1			
Guizhou					1			
Yunnan					1			
Shaanxi	1							
Gansu					1			
2+								
province								
Hebei	1				1	2		
Shanxi						1		
Inner Mongolia	2	1			1			
Liaoning	1				2			
Jilin		1						
Heilongjiang	1							
Jiangsu	1					1		
Zhejiang					2	2		
Anhui					1	3	1	
Fujian						3		
Jiangxi	1	1			1	4		
Shandong	2	2			2	5	1	
Henan			1		1	6		
Hubei	1				1	3		
Hunan						1		
Guangdong	1	3				14	3	
Guangxi						2		
Chongqing					2			
Sichuan					4		3	
Guizhou						1		
Yunnan					1			
Shaanxi						3	1	
Gansu						2		
Qinghai							2	
Xinjiang	1							
Total								
province								
Total				37				136

Table D.6: Policy area change across sampled counties (community head response), first change (conditional on changing)

Notes: The results in the table conform with anecdotal evidence that at the first policy change, rural areas moved from a strict one-child policy to a more relaxed rule. Similarly, communities and villages with an initial 1.5 or 2+ child policy reverted to a stricter policy type at the first change.

I show average characteristics of the sample of mothers by the total number of children ever born in Table D.7. Since this paper is investigating the impact of fertility on elderly cognition, mothers with a different number of children should differ in the cognition outcome. Birth cohort is strongly related to fertility as previously shown in Figure 2. Therefore, I only report summary statistics for one group of cohorts who were impacted by the OCP for the entire duration of their childbearing years (I choose cohorts 1960 to 1969 for this). Mothers who during their fertility window only have one child (all women in the sample are older than 45 and thus can be considered to have completed their fertility window) are less likely to work, and have a higher probability of having a male firstborn. A higher share of these mothers have a urban hukou (56%) and live in an urban community (35%). They are less likely to be low-educated, and have a better standardized cognition score (1.06 compared to 0.82). Albeit visible, these differences do not constitute any causal interpretation of the impact of family size on cognition, which the remaining part of this paper is investigating.

Table D.7: Mother characteristics by number of children

	# of children	
	1 4,181 (26.9%)	2+ children 11,384 (73.1%)
age	49.95 (3.13)	50.25 (3.18)
male	0.00 (0.00)	0.00 (0.00)
Cognition (Z-score)	0.61 (0.87)	0.13 (0.96)
Cognition Total	16.91 (4.67)	14.36 (5.14)
Cognitive Impairment (cutoff)	0.12 (0.33)	0.27 (0.44)
Cognitive Impairment (Bottom 20%)	0.06 (0.24)	0.18 (0.38)
working	0.69 (0.46)	0.76 (0.43)
# of living children	0.99 (0.11)	2.38 (0.81)
has at least 2 children	0.00 (0.00)	0.75 (0.43)
number of living daughters	0.35 (0.48)	1.17 (0.88)
number of living sons	0.64 (0.48)	1.20 (0.74)
any child in same city/county	0.67 (0.47)	0.78 (0.42)
any in-person contact	0.70 (0.46)	0.76 (0.43)
net asset quartile	3.08 (1.02)	2.72 (1.02)
rural hukou	0.55 (0.50)	0.88 (0.32)
education level		
education: primary	2,954 (70.7%)	10,411 (91.5%)
education: college	1,227 (29.3%)	969 (8.5%)
cohort group (10-year)		
1960	4,181 (100.0%)	11,384 (100.0%)

Table D.8: Mother characteristics by one-child policy area

		Policy Han (2011)		
		1	1.5	2+
		4,964 (40.0%)	6,124 (49.3%)	1,332 (10.7%)
Cohorts 1930-1949	age	71.38 (5.59)	71.23 (5.84)	71.66 (5.65)
	male	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
	Cognition (Z-score)	-0.31 (1.09)	-0.69 (1.00)	-0.84 (1.04)
	Cognition Total	11.99 (5.84)	9.96 (5.37)	9.13 (5.60)
	Cognitive Impairment (cutoff)	0.45 (0.50)	0.60 (0.49)	0.66 (0.47)
	Cognitive Impairment (Bottom 20%)	0.35 (0.48)	0.47 (0.50)	0.55 (0.50)
	working	0.29 (0.45)	0.39 (0.49)	0.46 (0.50)
	# of living children	3.35 (1.49)	3.89 (1.56)	3.95 (1.73)
	has at least 2 children	0.72 (0.45)	0.83 (0.38)	0.81 (0.39)
	number of living daughters	1.61 (1.20)	1.84 (1.29)	1.88 (1.38)
	number of living sons	1.73 (1.12)	2.05 (1.17)	2.06 (1.23)
	any child in same city/county	0.84 (0.37)	0.89 (0.32)	0.91 (0.28)
	any in-person contact	0.79 (0.41)	0.82 (0.38)	0.87 (0.34)
	net asset quartile	2.40 (1.19)	1.88 (1.01)	1.94 (0.94)
	education level			
	education: primary	4,416 (89.0%)	6,048 (98.8%)	1,324 (99.4%)
	education: college	548 (11.0%)	72 (1.2%)	8 (0.6%)
	cohort group (10-year)			
	1930	1,508 (30.4%)	1,800 (29.4%)	420 (31.5%)
	1940	3,456 (69.6%)	4,324 (70.6%)	912 (68.5%)

Appendix E: Alternative Regressions

I examine whether instrument relevance is sensitive to the inclusion of controls that may confound the first stage. Since household wealth could be correlated with both the ability to pay fines and fertility decisions, I progressively add marital status, health insurance coverage, and household asset quartiles (averaged over 2011–2018) as controls. I also test robustness to quadratic cohort trends. Table E.2 presents these first-stage estimates. The coefficient on firstborn girl ranges from 0.039 (s.e. 0.014) in the baseline to 0.033 (s.e. 0.015) in the most saturated specification, while the fine coefficient ranges from -0.121 to -0.129 , remaining significant at the 1% level throughout. The F-statistic declines from 136.24 in the baseline to 100.70 with all additional controls, and rises slightly to 102.06 with quadratic cohort trends, confirming strong instrument relevance across all specifications.

Table E.1: First Stage Specification: Alternative Instrument Specification

	(1) Main	(2) Alternative
girl	0.039*** (0.014)	0.044*** (0.014)
(1-girl)×fine	-0.121*** (0.013)	
fine	.	-0.147*** (0.014)
girl×fine	.	0.112*** (0.015)
Controls	Yes	Yes
Wave FE	Yes	Yes
County FE	Yes	Yes
Cohort trend	Linear	Linear
Clustering	Individual	Individual
Instr. F	136.24	119.76
N	11062	11062

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

Notes: Column (1) shows the main specification using $(1 - \text{girl}) \times \text{fine}$ as an instrument, which restricts fine effects to boy-first families. Column (2) shows the alternative specification using girl, fine, and $\text{girl} \times \text{fine}$ separately, which allows differential fine effects by firstborn gender. Both specifications use the same cognition sample (N=11,062). The alternative specification confirms that the fine deterrence effect is concentrated on families with a firstborn son: the fine effect for boys is -0.147 , while for girls it is $-0.147 + 0.112 = -0.035$ (76% weaker). The Wald test confirms the fine effect for girls is statistically significant but economically small. The coefficient difference between specifications (-0.121 vs -0.147) reflects the imposed restriction in the main specification versus the unrestricted interaction model.

Table E.2: First Stage Sensitivity and Robustness

	(1)	(2)	(3)	(4)	(5)
Firstborn girl	0.039*** (0.014)	0.039*** (0.014)	0.034** (0.014)	0.034** (0.015)	0.033** (0.015)
Fine	-0.121*** (0.013)	-0.121*** (0.013)	-0.125*** (0.014)	-0.127*** (0.015)	-0.129*** (0.015)
Cohort Trend	Linear	Linear	Linear	Linear	Quadratic
County FE	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes
Marital Status	No	Yes	Yes	Yes	Yes
Health Ins.	No	No	Yes	Yes	Yes
Asset Quartile	No	No	No	Yes	Yes
Inst. F-stat	136.24	136.37	126.80	100.70	102.06
N	11062	11062	10630	9044	9044

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

Sources of Identifying Variation in the Fine Instrument

Fines varied across provinces and over time. To determine whether identification comes from regional differences in fine levels or from temporal variation across birth cohorts within provinces, I compare first-stage regressions with and without province fixed effects.

Table E.3 presents the results. Column (1) estimates the first stage without province FE, exploiting both between-province and within-province variation. Column (2) adds province FE, which absorbs all time-invariant differences between provinces. Province FE eliminate cross-province variation in fine levels, leaving only within-province variation across birth cohorts. Column (3) shows the main specification with county FE.

The F-statistic is 129.3 with province FE (Column 2) compared to 113.6 without province FE (Column 1). The within R^2 is 0.118 in the province FE specification and 0.121 in the main specification. The instruments explain the same amount of fertility variation whether or not between-province differences are absorbed. This indicates that identification comes from within-province temporal variation across cohorts, not from cross-province differences in policy stringency.

This result is consistent with how fines evolved during the OCP period. While fines were set at the province level, enforcement changed substantially over time. Women born in the 1950s faced different fine schedules during their childbearing years than women born in the 1960s or 1970s, even within the same province. The IV estimates are therefore identified from these temporal policy changes affecting different cohorts, rather than from regional differences in fine levels.

Table E.3: Regional vs Temporal Variation in the Fine Instrument

	(1) Total Variation	(2) Within-Province	(3) Main
girl	0.026* (0.014)	0.038*** (0.014)	0.039*** (0.014)
(1-girl) $\times$ fine	-0.142*** (0.015)	-0.126*** (0.013)	-0.121*** (0.013)
Fixed Effects	None	Province	County + Wave
Variation	Regional + Temporal	Temporal Only	Temporal Only
Clustering	Individual	Individual	Individual
Instr. F	113.57	129.26	136.24
Within R^2	NA	0.118	0.121
N	11062	11062	11062

Standard errors clustered at individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age, age², education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

SUPPLEMENTARY MATERIAL

Table S.1: Codebook of main variables used in the analysis

Variable	Description	Min	Max	N
childh	Number of children	0	15	35,931
hachgendr1	Sex of firstborn child (1=Male, 0=Female)	0	1	46,322
age	Age of respondent	45	88	35,931
rabyear	Year of birth	1930	1979	46,322
raeduc1	Education level (categorical)	0	1	46,304
workr	Labor supply (1=Working, 0=Not working)	0	1	35,180
fcanyh	Received any financial transfer from children (1=Yes, 0=No)	0	1	34,759
fcamth	Financial transfer amount received from children (Yuan)	0.00	16600.00	34,701
coresdh	Any child co-resides in same city/county (1=Yes, 0=No)	0	1	35,672
fine_m2	Fine rate (multiples of annual income)	0.00	5.00	45,887
ocp_pol3c	Policy type (1=1-child, 2=1.5-child, 3=2/3-child)	1	3	45,696
cog_std	Standardized cognition score	-2.54	2.86	28,242
ruralcomm	Rural community (1=Rural, 0=Urban)	0	1	46,322

Notes: Sample includes mothers born between 1930 and 1974 in the CHARLS sample.

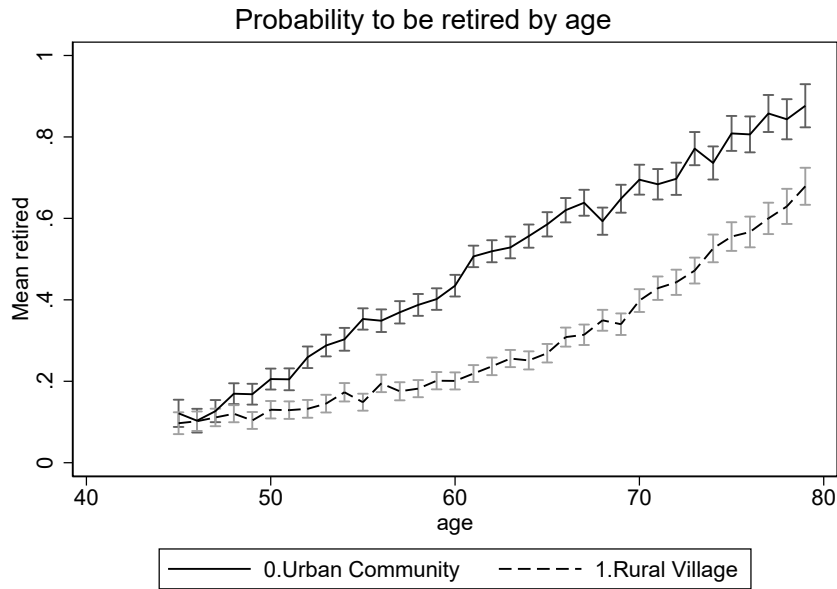


Figure S.1: Retirement probability by age.

	Policy rule		
	1	1.5	2
province			
Beijing	1		
Tianjin	1		
Hebei		1	
Shanxi		1	
Inner Mongolia		1	
Liaoning		1	
Jilin		1	
Heilongjiang		1	
Shanghai	1		
Jiangsu	1		
Zhejiang		1	
Anhui		1	
Fujian		1	
Jiangxi	1		
Shandong		1	
Henan		1	
Hubei		1	
Hunan		1	
Guangdong		1	
Guangxi		1	
Hainan			1
Chongqing	1		
Sichuan	1		
Guizhou		1	
Yunnan			1
Tibet		1	
Shaanxi		1	
Gansu		1	
Qinghai			1
Ningxia			1
Xinjiang			1

Table S.2: Policy area by province

Notes: The table shows which provinces implemented predominantly which one-child-policy rule, according to the classification by Ebenstein (2010). Hainan and Ningxia, and the remote Western regions of Xinjiang, Qinghai and Yunnan mostly had a two-child policy regime in place.

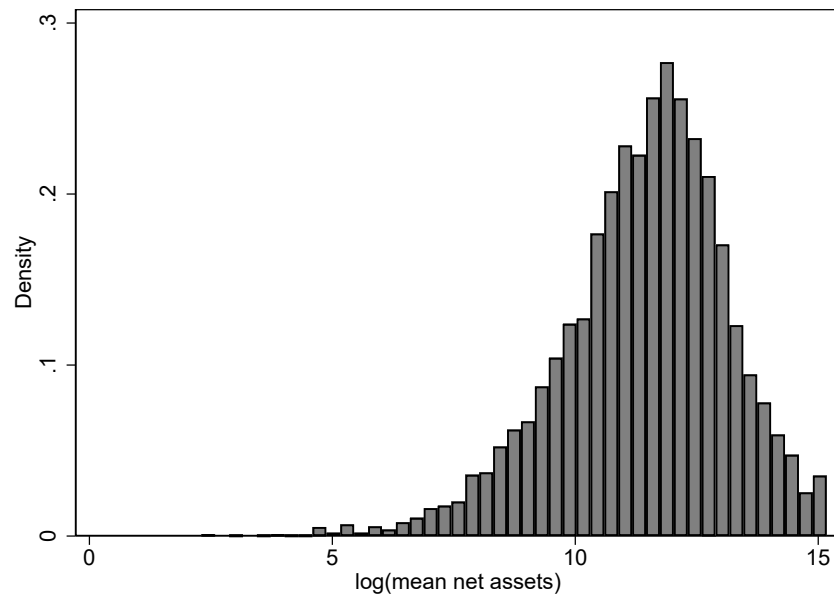


Figure S.1: Distribution of wealth.

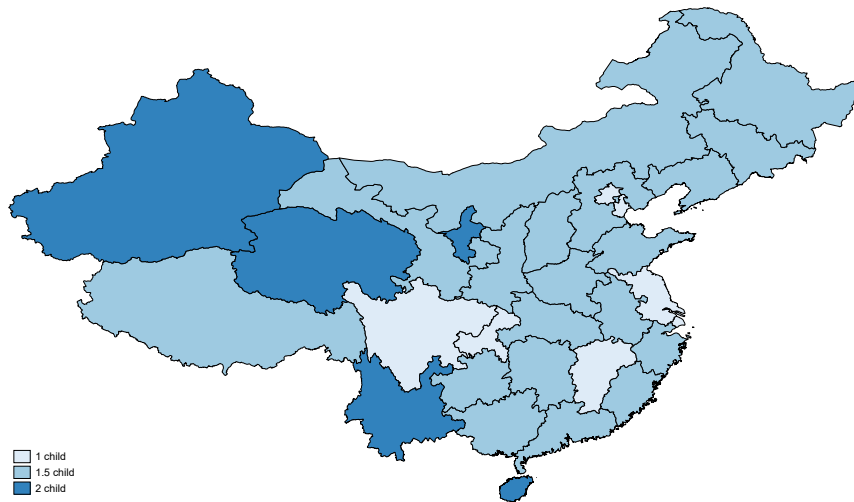


Figure S.2: Distribution of 1, 1.5 and 2-child regimes

Notes: The figure illustrates the distribution of 1, 1.5 and 2-child policies across provinces.

Table S.1: Between IV: All Outcomes (Robustness)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Cognition (std)	Cognition (total)	Cog. impairment	Working	Weeks worked	Hours/week	Lives near child	Any transfers	Transfers
F	-0.017 (0.147)	-0.092 (0.788)	-0.004 (0.066)	0.076 (0.056)	-1.038 (2.432)	-3.459 (3.616)	0.170*** (0.057)	0.688*** (0.067)	3451.694*** (882.315)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort trend	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear
Estimator	Between	Between	Between	Between	Between	Between	Between	Between	Between
KP F	110.51	110.51	110.51	122.23	124.40	125.49	120.60	123.43	123.59
N	4116	4116	4116	4306	3725	3715	4318	4324	4323

Robust standard errors (HC1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Controls: birth cohort (linear), age (mean), age² (mean), education dummies (primary vs. above).

Sample: mothers in 1.5-child policy areas.

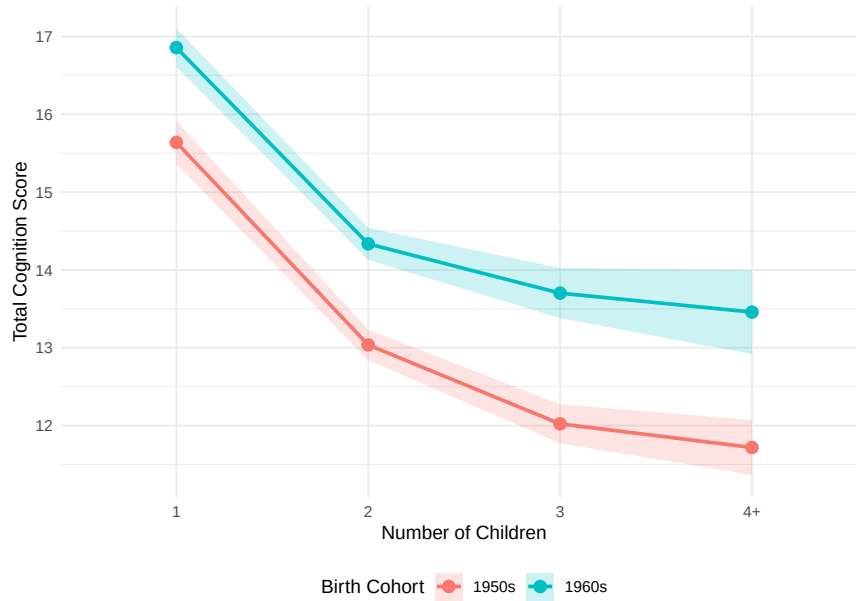


Figure S.3: Cognition Score by Fertility and Birth Cohort (Ages 50-60)

Notes: Mean cognition scores across fertility levels (1-4+ children) with separate lines for birth cohorts (1930s-early 1970s), restricted to ages 50-60 to control for age effects across cohorts. Shaded areas represent 95% confidence intervals. The 7-year observation window (2011-2018) limits cross-cohort comparisons at identical ages; future research can reevaluate as cohorts age. Sample: mothers born 1930-1974 observed at ages 50-60.

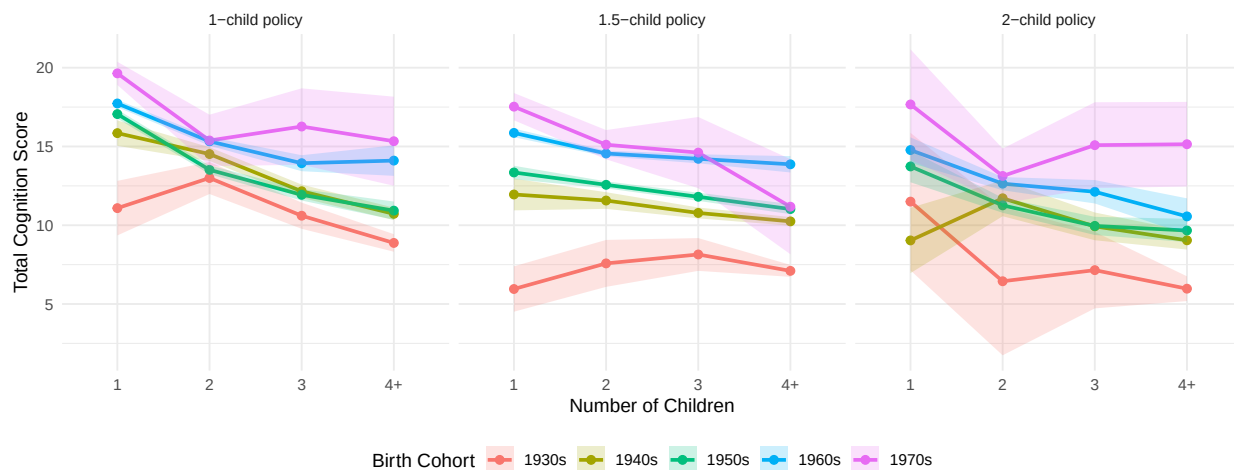


Figure S.4: Cognition Score by Fertility, Birth Cohort, and Policy Area

Notes: Mean cognition scores across fertility levels (1-4+ children) with separate lines for birth cohorts (1930s-early 1970s). Three subpanels show results by policy area (1-child, 1.5-child, 2-child). Shaded areas represent 95% confidence intervals. Sample: all mothers born 1930-1974.

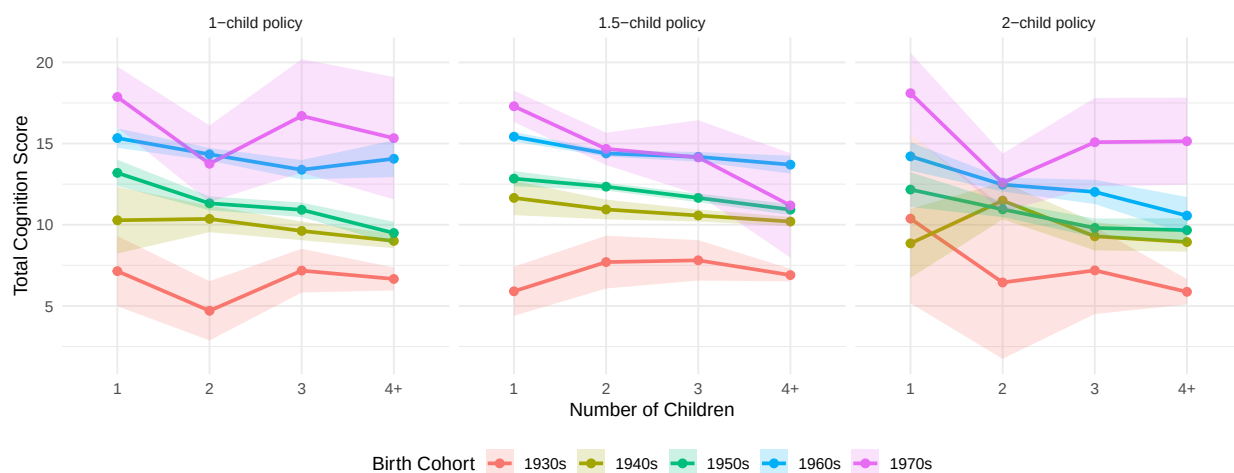


Figure S.5: Cognition Score by Fertility, Birth Cohort, and Policy Area (Rural Areas)

Notes: Mean cognition scores across fertility levels (1-4+ children) with separate lines for birth cohorts (1930s-early 1970s). Three subpanels show results by policy area (1-child, 1.5-child, 2-child). Shaded areas represent 95% confidence intervals. Sample: mothers in rural areas born 1930-1974.

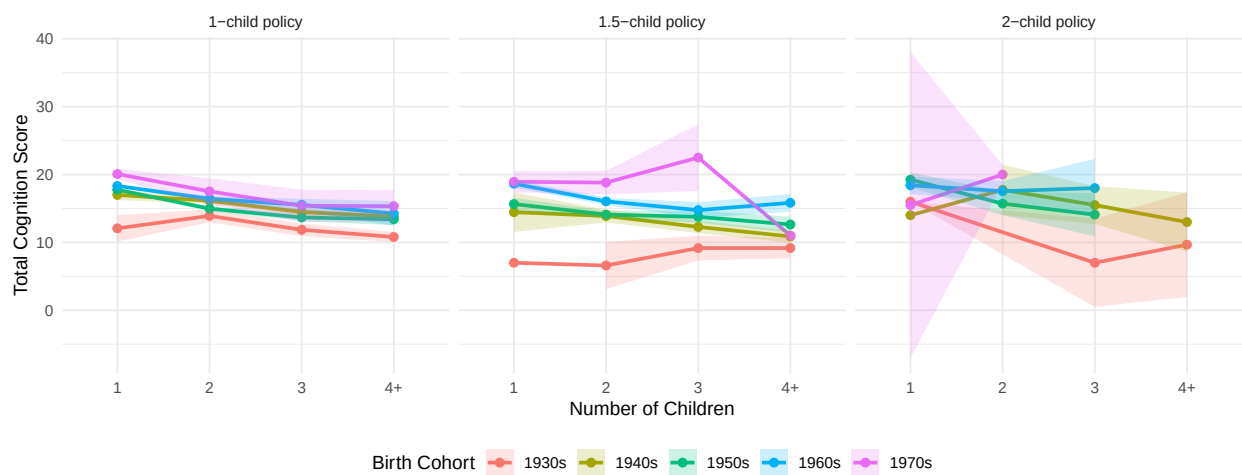


Figure S.6: Cognition Score by Fertility, Birth Cohort, and Policy Area (Urban Areas)

Notes: Mean cognition scores across fertility levels (1-4+ children) with separate lines for birth cohorts (1930s-early 1970s). Three subpanels show results by policy area (1-child, 1.5-child, 2-child). Shaded areas represent 95% confidence intervals. Sample: mothers in urban areas born 1930-1974.